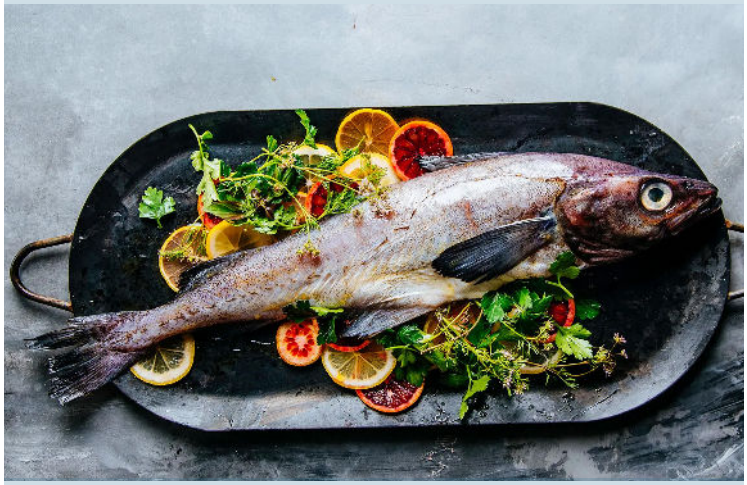

The Economic Contribution of the Alaska Pollock Fishery

Prepared for

The Alaska Pollock Fishery Alliance

September 2025

Northern Economics

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Abbreviations

AFA	American Fisheries Act
APFA	Alaska Pollock Fishery Association
BSAI	Bering Sea and Aleutian Islands
CDQ	Community Development Quota
GOA	Gulf of Alaska
IMPLAN	Impact Analysis for Planning
I-O	Input Output
NOAA	National Oceanic and Atmospheric Administration
O&M	Operations and Maintenance

Executive Summary

In December of 2024, Northern Economics was contracted by the Alaska Pollock Fishery Alliance to quantify the economic contribution of the Alaska pollock fisheries to the Alaska state economy as well as the economies of Washington, Oregon, and the rest of the United States. In this report we summarize the findings of the study, including total employment, income, and business sales (or economic output) as a result of Alaska pollock fishery harvesting and processing activities in 2023. The estimates of the economic benefits of the industry are based on IMPLAN model results using information obtained from a survey of Alaska pollock harvesting and processing vessels that participated in, or processed products from, either the Bering Sea and Aleutian Island or Gulf of Alaska (GOA) pollock fisheries in 2023. It also captures impacts associated with the employment and spending of companies and Western Alaska Community Quota Development (CDQ) groups who owned or co-owned Alaska pollock vessels or processors or received Alaska pollock quota royalties in 2023.

Study Coverage Rates

In total, in comparison to reported survey harvest or purchased volumes against American Fisheries Act (AFA) cooperative reports[1] and NOAA Fisheries Groundfish Economic Stock Assessment and Fishery Evaluation[2] reported harvest volumes (NOAA Fisheries 2024), the survey results captured approximately 35% of catcher vessel harvest, 77% of catcher-processor harvest, 100% of mothership purchases, and 60% of shorebased processor purchases of Alaska pollock in 2023.

Northern Economics also produced a survey of fishing companies to capture spending and employment not directly associated with harvesting or processing fish onboard vessels or in processing plants (called survey of "other employment and spending"). This was intended to document other employment and spending not captured in the individual sector surveys (e.g., headquarters offices, corporate offices, and regional offices not directly associated with harvesting or processing operations). In total, companies associated with 36% of the total harvesting and purchase volume responded to the survey, noting that not all participants may not have had this type of spending to report, so this may be an underestimate.

Finally, Northern Economics also interviewed Western Alaska CDQ groups in order to understand how revenue associated with the direct operation of Alaska pollock fishing vessels/processors or through Alaska pollock quota royalties and joint ventures support CDQ employment and spending.

[1] *https://www.npfmc.org/cooperative-reporting/*

[2] *https://www.fisheries.noaa.gov/alaska/ecosystems/economic-status-reports-gulf-alaska-and-bering-sea-aleutian-islands*

Through this effort, information from five of the six CDQ groups was collected and used to refine and supplement publicly available information from annual CDQ reports.

The number of vessels, processors, companies, and CDQ groups who participated in the study is summarized below in ES Table 1.

ES Table 1. Number of Survey Responses by Sector or Type of Survey

Sector/Type of Survey	Number of Surveys or Interviews
Catcher Vessels	33
Motherships	3
Catcher-Processors	10
Shorebased Processors	6
Other Employment and Spending	12
CDQ Employment and Spending	5

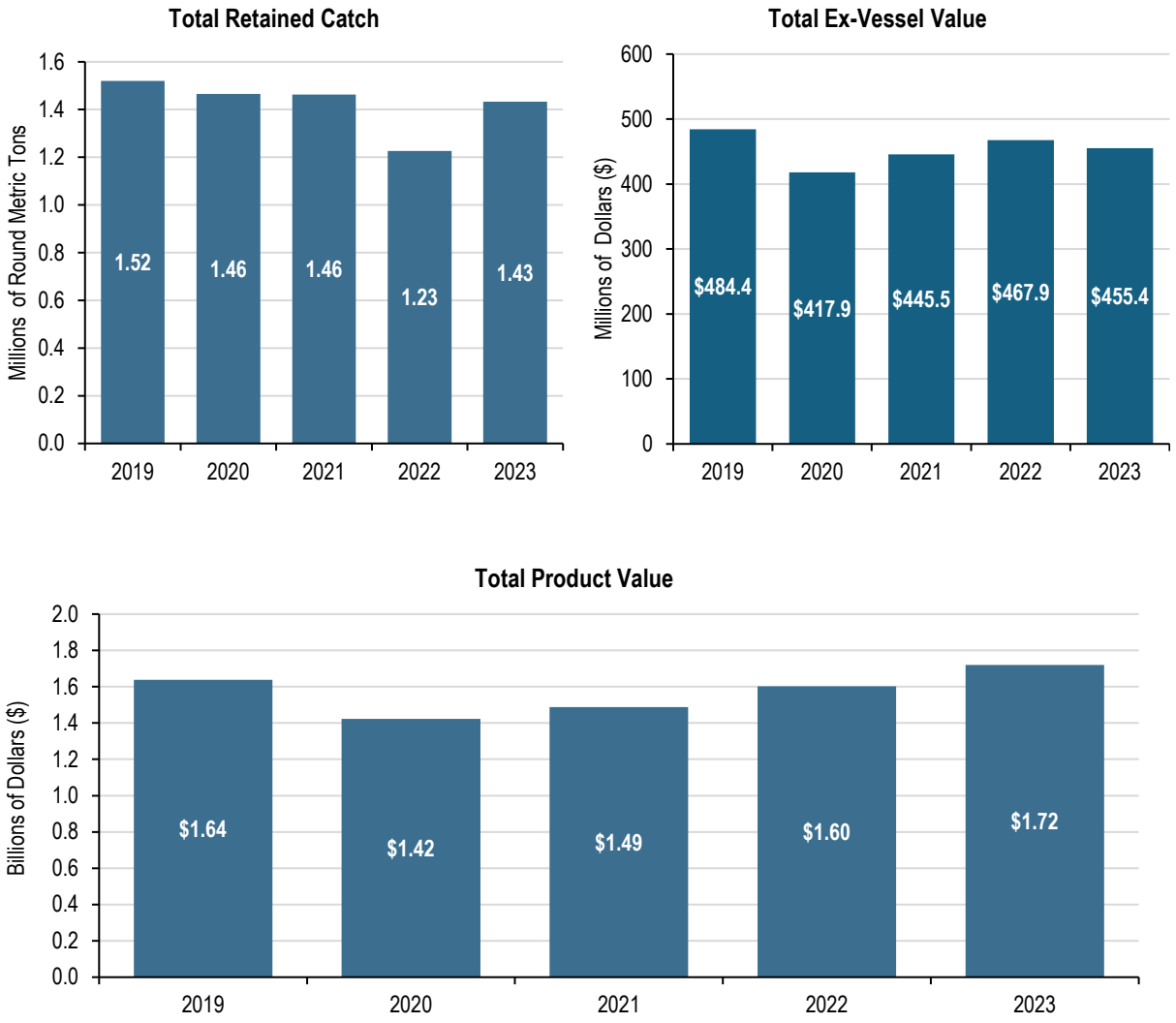
Source: Northern Economics Analysis

Northern Economics used survey responses within sectors to estimate total spending and employment supported by all Alaska pollock fishery activity in 2023. These results are summarized in the following sections, along with a fiscal analysis of estimated Alaska tax revenue generated by the fishery in the same year.

2023 Fishery Landings and Revenue

In 2023, 1.43 million metric tons of Alaska pollock were caught across all sectors and regions at an ex-vessel value of over $455 million (ES Figure 1). The estimated total Alaska pollock product value in 2023 is $1.7 billion (ES Figure 1). While retained catch and ex-vessel revenue in 2023 remained lower than in 2019, the estimated total product value was at a five-year high.

ES Figure 1. Total Alaska Pollock Catch and Value 2019-2023

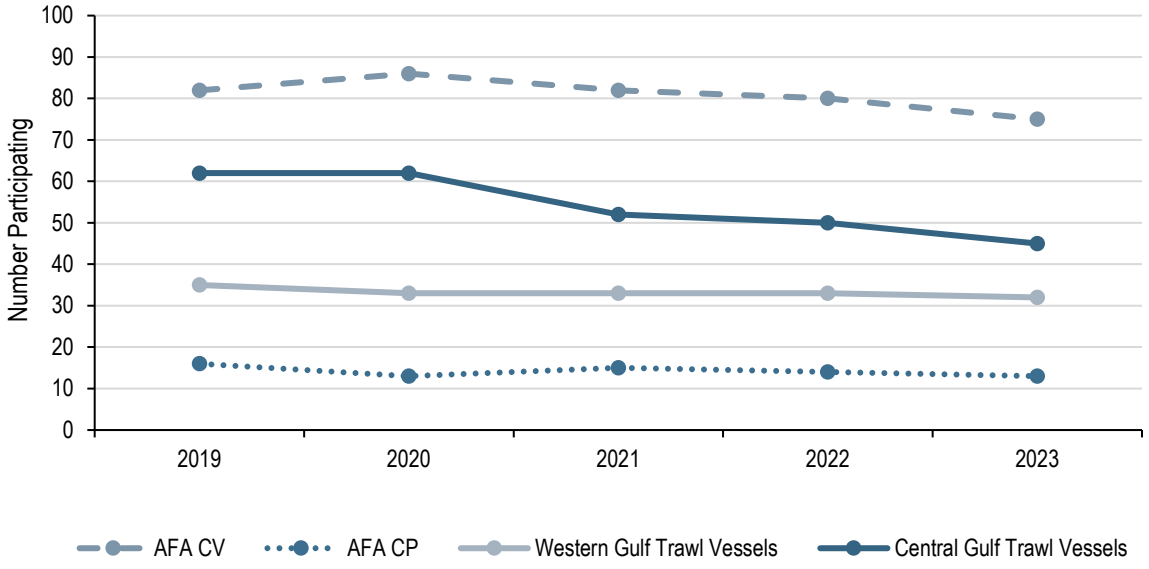


Notes: Upper Left: Retained catch does not include discarded catch. Only includes catch counted against federal TACs. Upper Right: Values are not adjusted for inflation. The value added by at-sea processing is not included in these estimates of ex-vessel value. Bottom: Values have not been adjusted for inflation. Values are calculated using NOAA Fisheries estimates of product value per mt of retained catch, NOAA Fisheries estimates of retained catch volume for CP and CV sectors by area, and MS reported landing volumes from annual AFA cooperative reports.

Source: NOAA Fisheries 2024; AFA cooperative reports, and Northern Economics Analysis

ES Figure 2 shows the number of vessels that participated in the AFA and GOA groundfish fisheries from 2019 to 2023. During this period, there were declines in participation in the number of active catcher vessels and catcher processors. ES Figure 3 shows the number of processors participating in these fisheries from 2019 to 2023. All processor types except Western Gulf processors declined over that period. In 2023, there were 12 Western Gulf processors participating compared to 10 in 2019. AFA processors that were not catcher processors (i.e., motherships and shorebased processors) experienced the largest decline from 2019 to 2023, with only half as many participants in 2023 compared to 2019.

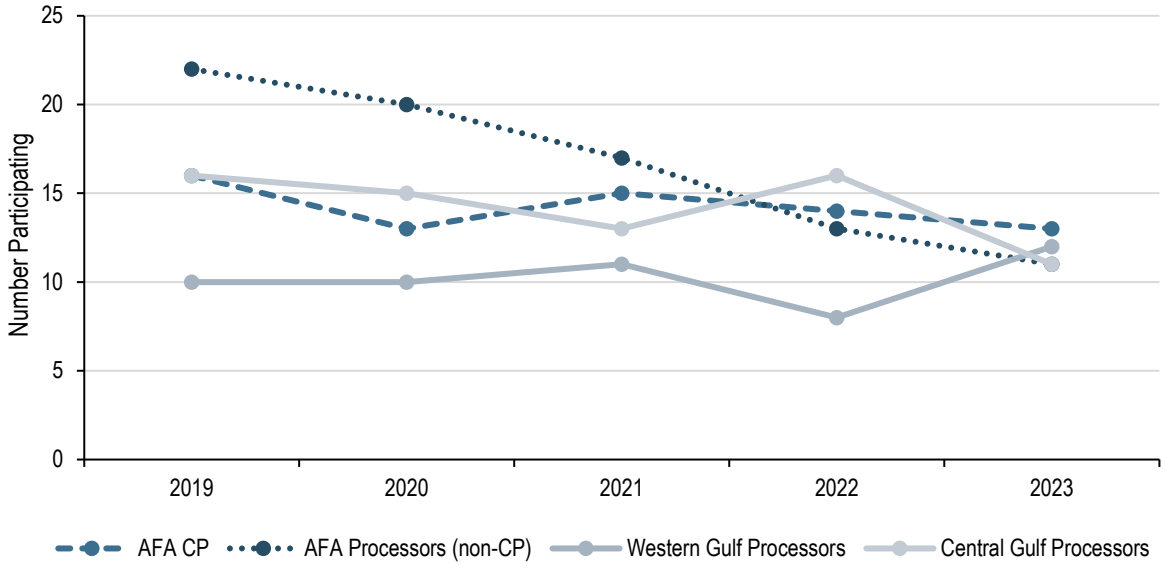
ES Figure 2. AFA and GOA Groundfish Fishery Vessel Participation 2019–2023



Note: Number of Central and Western Gulf of Alaska vessels represent all GOA groundfish vessels, not just vessels that fished for Alaska pollock.

Source: NOAA Fisheries 2024

ES Figure 3. AFA and GOA Groundfish Fishery Processor Participation 2019–2023



Note: Number of Central and Western Gulf of Alaska vessels represent all GOA groundfish vessels, not just vessels that fished for Alaska pollock.

Source: NOAA Fisheries 2024

Economic Contribution

The economic contribution of the Alaska pollock fishery was measured in terms of employment, labor income, and economic output (or business sales) generated in Alaska, Washington, Oregon, and the rest of the U.S. The total economic contribution of the fishery is comprised of direct effects and multiplier effects, which include indirect and induced effects. The direct effects result from the industry's spending on operations and maintenance (labor and non-labor) and capital spending. The multiplier effects are the stimulus effects generated in the various states' economies that result from the spin-off spending as other businesses that support the industry's annual operations and capital spending buy goods and services from suppliers within the state (indirect effects), and as employees spend their income on food, utilities, clothing, and other personal expenditures in the areas where they reside (induced effects).

For this study, the direct effects are associated with the level of employment and spending directly attributable to the Alaska pollock fishery in 2023. This includes the total reported amount of spending and employment (individuals by place of residence), adjusted for the proportion directly attributable to the Alaska pollock fishery and Northern Economics' estimates of employment and spending for the rest of the sector, if the survey sampled less than 100% of vessels, processors, or companies in a sector. The direct non-labor operations and maintenance (O&M) and capital spending by all the Alaska pollock fishery sectors—catcher vessels, catcher processors, motherships, shorebased processors, CDQs, and their workers' spending of wages—were used as inputs in the IMPLAN[3] model to quantify the indirect and induced effects (multiplier effects) in each of the states where the expenditures occurred (or where employees reside). The spending on goods and equipment purchased from outside the U.S. (here referred to as the 'rest of the world') and the spending of wages by workers who were not residents of the U.S. were considered leakages and therefore did not generate multiplier effects in the U.S. economy.

Employment

ES Table 2 shows employment effects of the Alaska pollock industry in 2023. The industry supported an estimated 16,821 direct, indirect, and induced jobs in the U.S.

Direct employment is the estimated number of employees supported directly by the Alaska pollock industry. Indirect employment is the estimated number of jobs supported in sectors that provided goods and services to the Alaska pollock industry for annual operations and maintenance as well as capital spending in 2023. Induced employment includes jobs supported in sectors that provide goods and services to households, through the spending of wages by the individuals who worked in the

[3] The IMPLAN model is an input-output (I-O) model that measures how many times a dollar is re-spent in, or "ripples" through the economic region. The I-O model yields multipliers that are used to calculate the indirect and induced economic effects in terms of jobs, labor income, and business sales generated within a specified region per dollar of industry spending on various types of goods and services.

Alaska pollock fish harvesting, processing, and seafood company sectors as well as those employed by CDQs. These induced effects were generated in the areas where the workers live.

In 2023, it is estimated that the industry directly employed 8,277 workers from various states nationwide—31% were Alaskans, 23% were from Washington, 3% were from Oregon, and 32% were from other states. An additional 1,089 workers were residents of other countries (12%).

In addition to direct jobs, spending on goods and services by the Alaska pollock industry supported an additional 8,544 indirect and induced jobs nationwide. About 76% of these additional jobs were in Alaska (40%), Washington (34%), and Oregon (2%).

In total, the greatest employment effects are in Alaska, where total jobs in 2023 were estimated to be 6,318. Alaska's direct and indirect employment exceed the levels in Washington, particularly indirect jobs. Alaska had over 900 more indirect jobs supported by the Alaska pollock industry than Washington, reflecting that spending on goods and services is higher in Alaska. Total employment effects were estimated to be 5,045 jobs in Washington, and 421 jobs in Oregon. Additionally, about 70% of the Alaska pollock industry effects nationwide in 2023 were in Alaska, Washington, and Oregon.

ES Table 2. Direct, Indirect, and Induced Employment by Location

	Employment			
Location	Direct	Indirect	Induced	Total
Alaska	2,869	2,386	1,064	6,318
Washington	2,168	1,457	1,419	5,045
Oregon	265	36	121	421
AK, WA, OR Subtotal	**5,302**	**3,878**	**2,604**	**11,784**
Rest of U.S.	2,975	322	1,739	5,036
Nationwide Total	**8,277**	**4,201**	**4,343**	**16,821**
Rest of the World	1,089	0	0	1,089
Global Total	**9,366**	**4,201**	**4,343**	**17,910**

Source: Northern Economics Analysis

Labor Income

ES Table 3 shows the labor income effects of the Alaska pollock industry in 2023. The Alaska pollock industry paid about $459 million in direct labor income (wages and benefits) nationwide; these were the payments to individuals who were directly employed by the Alaska pollock industry in 2023. The estimated additional indirect and induced labor income generated nationwide amounted to about $694 million. Combined, the Alaska pollock industry generated an estimated $1.15 billion in labor income in 2023.

The direct labor income is comprised of wages, salaries, and benefits paid to the individuals directly employed by the 2023 Alaska pollock industry. Indirect income effects include wages paid for the jobs in sectors that provided goods and services to the Alaska pollock industry for annual operations and maintenance as well as capital spending (the indirect jobs supported in the *Employment* section above). Induced income effects are the wages associated with the jobs supported by the spending of the Alaska fishing industry's wages (i.e., wages of crew, processing workers, and other employees of fishing companies supported by the Alaska pollock industry).

Washington experienced the largest labor income effects of the three states analyzed individually ($450 million), even though Alaska had higher employment effects than Washington, reflecting higher average labor payments per employee. Total labor income effects in Alaska were estimated to be $337 million. Collectively Alaska, Washington, and Oregon accounted for 70.3% of total labor income effects of the Alaska pollock industry in 2023, while labor income effects in other US states and other countries accounted for 26.4% and 3.3%, respectively.

ES Table 3. 2023 Direct, Indirect, and Induced Labor Income by Location

	Labor Income			
Location	Direct	Indirect	Induced	Total
Alaska	$102,913,225	$165,626,096	$68,360,728	$336,900,050
Washington	$171,675,711	$170,519,356	$107,756,408	$449,951,474
Oregon	$17,454,328	$22,232,360	$12,427,625	$52,114,313
AK, WA, OR Subtotal	**$292,043,264**	**$358,377,812**	**$188,544,761**	**$838,965,837**
Rest of U.S.	$167,132,990	$27,241,557	$120,017,217	$314,391,765
Nationwide Total	**$459,176,254**	**$385,619,369**	**$308,561,979**	**$1,153,357,602**
Rest of the World	$39,363,570	$0	$0	$39,363,570
Global Total	**$498,539,824**	**$385,619,369**	**$308,561,979**	**$1,192,721,171**

Source: Northern Economics Analysis

Economic Output (Total Business Sales)

For this study, the economic output measures the total business sales generated by the Alaska pollock industry in each state as a result of industry-wide spending in 2023. The estimated total business sales generated by the Alaska pollock industry in the U.S. in 2023 amounted to about $2.54 billion, with $1.27 billion in direct economic output and $1.46 billion in indirect and induced economic output. The direct economic output is the total estimated direct spending by the Alaska pollock industry for labor and non-labor operations and maintenance, plus capital spending. Indirect economic output effects are the amount of total business activity generated in various supplier sectors resulting from demand for goods and services by the Alaska pollock industry's operations in 2023. Induced economic output effects are the amount of business sales generated as a result of workers' spending of wages.

ES Table 4 shows the estimated economic output by location of the Alaska pollock industry in 2023. Due to higher labor income effects discussed previously resulting in higher induced effects, Washington experienced the largest economic output effects of the Alaska pollock industry of the three states analyzed individually at $961 million, with Alaska at $832 million. Alaska, Washington, and Oregon combined accounted for about 75% of the nationwide economic output of the Alaska pollock industry in 2023.

ES Table 4. Direct, Indirect, and Induced Business Sales (Economic Output) by Location

	Business Sales (Economic Output)			
Location	Direct	Indirect	Induced	Total
Alaska	$410,181,534	$231,640,814	$190,542,512	$832,364,860
Washington	$407,662,842	$228,250,766	$324,724,405	$960,638,014
Oregon	$50,223,946	$31,666,238	$34,977,222	$116,867,406
AK, WA, OR Subtotal	**$868,068,322**	**$491,557,818**	**$550,244,140**	**$1,909,870,280**
Rest of U.S.	$207,605,675	$47,220,244	$373,232,476	$628,058,396
Nationwide Total	**$1,075,673,998**	**$538,778,062**	**$923,476,616**	**$2,537,928,676**
Rest of the World	$51,712,603	$0	$0	$51,712,603
Global Total	**$1,127,386,600**	**$538,778,062**	**$923,476,616**	**$2,589,641,279**

Source: Northern Economics Analysis

Fiscal Impacts

Taxes paid to the state of Alaska, local boroughs, and cities can be estimated based on reported landed volumes and values of Alaska pollock in 2023 (NOAA Fisheries 2024) (ES Table 5). Across state fisheries business taxes, fishery resource landing taxes, any local or borough raw fish taxes, and other taxes (such as the Seafood Marketing Assessment tax), it is estimated that Alaska pollock generated $21 million in Alaska local, borough, and state tax revenue in 2023.

ES Table 5. Estimated 2023 Alaska Pollock Fishery Taxes in Alaska

Tax Type	2023 Estimate
State Fisheries Business Tax and Fishery Resource Landing Tax	$13,846,145
Local Raw Fish Taxes	$4,997,471
Other	$2,307,691
Total	**$21,151,307**

Note: Only Alaska state fishery taxes are estimated. Other taxes include seafood marketing assessment taxes.

Source: NOAA Fisheries 2024 and Northern Economics analysis

Introduction

In December of 2024, Northern Economics was contracted by the Alaska Pollock Fishery Association (APFA) to quantify the economic contribution of the Alaska pollock fisheries to the Alaska state economy as well as the economies of Washington, Oregon, and the rest of the United States. In this report we summarize the findings of the study, including total employment, income, and business sales (or economic output) as a result of Alaska pollock fishery harvesting, processing, ownership, and quota royalties in 2023. The estimates of the economic benefits of the industry are based on IMPLAN model results using information obtained from a survey of Alaska pollock harvesting and processing vessels and companies who participated in, or processed products from, either the Bering Sea and Aleutian Island (BSAI) or Gulf of Alaska (GOA) Alaska pollock fisheries in 2023. The model also includes estimates of economic benefits generated by Alaska pollock quota royalties and ownership of Alaska pollock vessels or processors by Western Alaska Community Development (CDQ) groups. These estimates were supported by an interview effort with CDQ groups to review and supplement information provided in their annual reports.

This report is organized as follows:

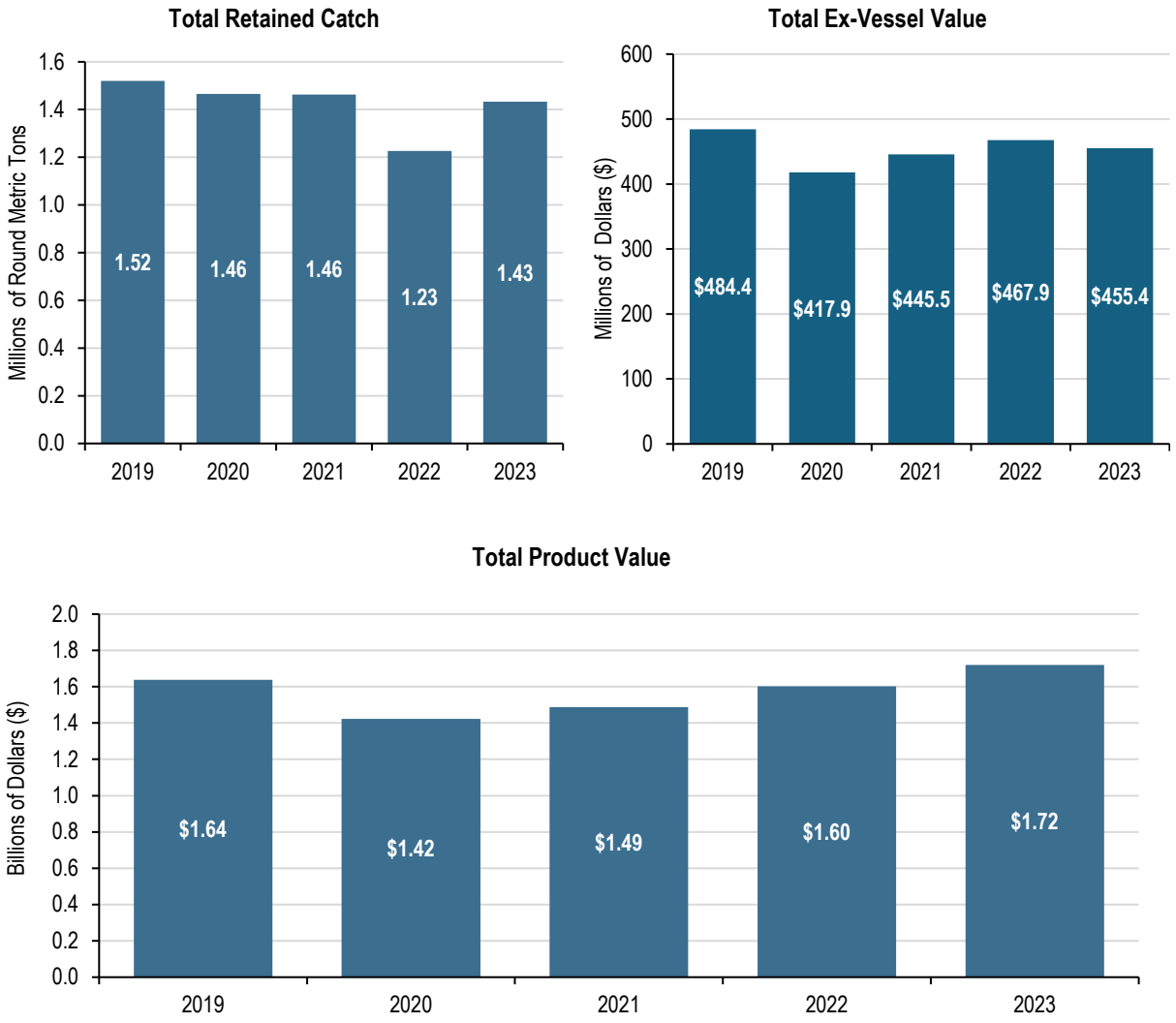
1. Background: Summary of the fishery in recent years
2. Results: Economic contribution and fiscal impact analyses
3. Conclusions: top level results and conclusions
4. Methods
5. References
6. Appendices

Background

The Alaska pollock fishery is managed through coordination between NOAA Fisheries and the North Pacific Fishery Management Council. The American Fisheries Act (AFA) rationalized the BSAI pollock fishery in 1998. The AFA greatly changed the Alaska pollock fishery in the Bering Sea by increasing efficiency, reducing bycatch, and utilizing a higher portion of catch among other effects (NPFMC 2025). Alaska pollock in the Bering Sea is harvested by shorebased (catcher vessels and shorebased processors) and at-sea sectors (catcher-processors and motherships). The Alaska pollock fishery in the GOA is managed as part of a Groundfish Fishery Management Plan which includes many other groundfish species that may be landed and targeted other than pollock. The total allowable catch for Alaska pollock in the Gulf of Alaska is allocated entirely to the shorebased sector across both the Western and Central Gulf (NPFMC,2024). In this report, the Alaska pollock fishery includes both the BSAI and GOA pollock fisheries and generally summarizes available data and economic effects stemming from ownership, harvesting, and processing activities across the BSAI and GOA management areas. It also includes economic effects stemming from economic activities supported by the revenue received by CDQ groups from quota royalties and ownership or co-ownership of Alaska pollock vessels or processors. The Western Alaska CDQ program allocates a percentage of all BSAI groundfish quotas to eligible communities. The six CDQ groups are allocated 10 percent of the Eastern Bering Sea pollock total allowable catch and most own or co-own Alaska pollock fishing vessels or catcher-processors (NOAA Fisheries 2025). Together, it is estimated that 30% of the fishery is owned by CDQ groups (Deakin, Alstrom, and Link 2024).

In 2023, 1.43 million metric tons of Alaska pollock were caught across all sectors and regions at an ex-vessel value of over $455 million (Figure 1). The estimated total Alaska pollock product value in 2023 is $1.7 billion (Figure 1).

Figure 1. Total Alaska Pollock Catch and Value 2019-2023



Notes: Upper left: Retained catch does not include discarded catch. Only includes catch counted against federal TACs. Upper right: Values are not adjusted for inflation. The value added by at-sea processing is not included in these estimates of ex-vessel value. Bottom: Values have not been adjusted for inflation. Values are calculated using NOAA Fisheries estimates of product value per mt of retained catch, NOAA Fisheries of retained catch volume for CP and CV sectors by area, and MS reported landing volumes from annual AFA cooperative reports.

Source: NOAA Fisheries 2024; AFA cooperative reports, and Northern Economics Analysis

Table 1. Total Alaska Pollock Catch and Value 2019-2023

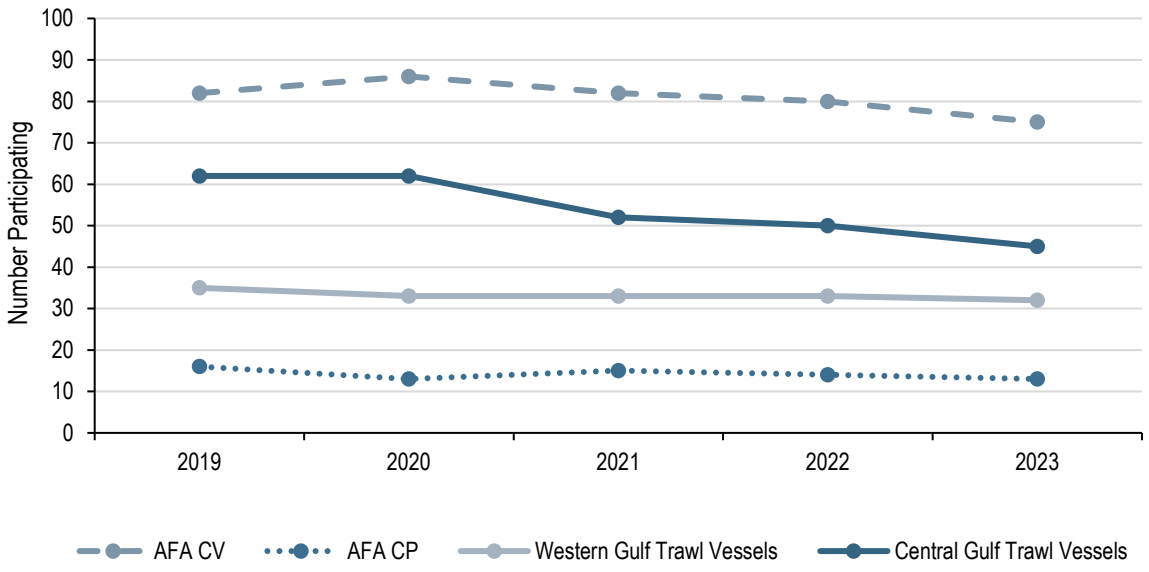
Year	Total Retained Catch (Millions of round mt)	Total Ex-vessel Value (Millions of Dollars)	Estimated Total Product Value (Billions of Dollars)
2019	1.5	$484.4	$1.6
2020	1.5	$417.9	$1.4
2021	1.5	$445.5	$1.5
2022	1.2	$467.9	$1.6
2023	1.4	$455.4	$1.7

Notes: Retained catch does not include discarded catch. Only includes catch counted against federal TACs. Ex-vessel and Product Values are not adjusted for inflation. The value added by at-sea processing is not included in these estimates of ex-vessel value. Bottom: Values are calculated using NOAA Fisheries estimates of product value per mt of retained catch, NOAA Fisheries estimates of retained catch volume for CP and CV sectors by area, and MS reported landing volumes from annual AFA cooperative reports.

Source: NOAA Fisheries 2024; AFA cooperative reports, and Northern Economics Analysis

Figure 2 shows the number of vessels that participated in the AFA and GOA groundfish fisheries from 2019 to 2023. During this period, there were declines in participation in the number of active catcher vessels and catcher processors. Figure 3 shows the number of processors participating in the AFA and GOA groundfish fisheries from 2019 to 2023. All processor types except Western Gulf processors declined over that period. In 2023, there were 12 Western Gulf processors participating compared to 10 in 2019. AFA Processors that were not catcher processors (i.e., motherships and shorebased processors) experienced the largest decline from 2019 to 2023, with only half as many participants in 2023 compared to 2019.

Figure 2. AFA and GOA Groundfish Fishery Vessel Participation 2019–2023



Note: Number of Central and Western Gulf of Alaska vessels represent all GOA groundfish vessels, not just vessels that fished for Alaska pollock.

Source: NOAA Fisheries 2024

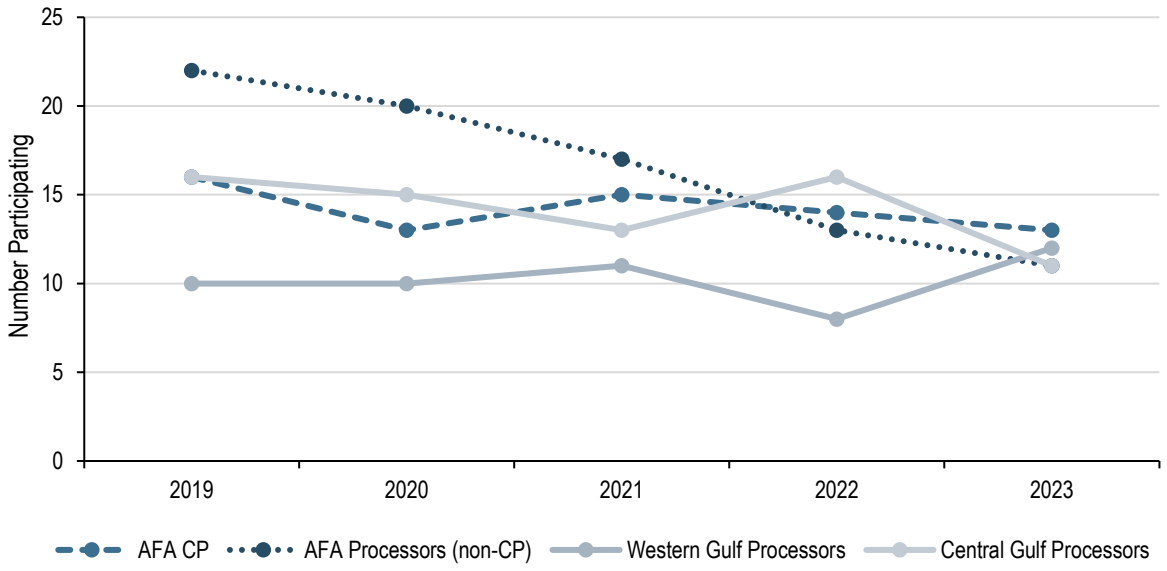
Table 2. AFA and GOA Groundfish Fishery Vessel Participation 2019–2023

Year	AFA CV	AFA CP	Western Gulf Trawl Vessels	Central Gulf Trawl Vessels
2019	82	16	35	62
2020	86	13	33	62
2021	82	15	33	52
2022	80	14	33	50
2023	75	13	32	45

Note: Number of Central and Western Gulf of Alaska vessels represent all GOA groundfish vessels, not just vessels that fished for Alaska pollock.

Source: NOAA Fisheries 2024

Figure 3. AFA and GOA Groundfish Fishery Processor Participation 2019–2023



Note: Number of Central and Western Gulf of Alaska vessels represent all GOA groundfish vessels, not just vessels that fished for Alaska pollock.

Source: NOAA Fisheries 2024

Table 3. AFA and GOA Groundfish Fishery Processor Participation 2019–2023

Year	AFA Processors (non-CP)	Central Gulf Processors	AFA CP	Western Gulf Processors
2019	22	16	16	10
2020	20	15	13	10
2021	17	13	15	11
2022	13	16	14	8
2023	11	11	13	12

Note: Number of Central and Western Gulf of Alaska vessels represent all GOA groundfish vessels, not just vessels that fished for Alaska pollock.

Source: NOAA Fisheries 2024

Results

Economic Effects

The economic contribution of the Alaska pollock fishery was measured in terms of employment, labor income, and economic output (or business sales) generated in Alaska, Washington, Oregon, and the rest of the U.S. The total economic contribution of the fishery is comprised of the direct effects and the multiplier effects, which include indirect and induced effects. The direct effects result from the industry's spending on operations and maintenance (labor and non-labor) and capital spending. The multiplier effects are the stimulus effects generated in the various states' economies that result from the spin-off spending as other businesses that support the industry's annual operations and capital spending buy goods and services from suppliers within the state (indirect effects), and as employees spend their income on food, utilities, clothing, and other personal expenditures in the areas where they reside (induced effects).

For this study, the direct effects are associated with the level of employment and spending directly attributable to the Alaska pollock fishing industry in 2023. This includes the total amount of spending and employment (individuals by place of residence) reported in the survey, adjusted for the proportion directly attributable to the Alaska pollock fishery, if appropriate, and Northern Economics' estimates of employment and spending for the rest of the sector, if the survey sampled less than 100% of vessels, processors, or companies in a sector (see *Methods* section for more information).

The direct non-labor operations and maintenance (O&M) and capital spending by all the Alaska pollock fishery sectors—catcher vessels, catcher processors, motherships, shorebased processors, CDQ groups, and their workers' spending of wages—were used as inputs in the IMPLAN[4] model to quantify the indirect and induced effects (multiplier effects) in each of the states where the expenditures occurred (or where employees reside). The spending on goods and equipment purchased from outside the U.S. (here referred to as the 'rest of the world') and the spending of wages by workers who were not residents of the U.S. were considered leakages and therefore did not generate multiplier effects in the U.S. economy.

Also note that in I-O models, multipliers measure only the backward linkages (or upstream effects). The analysis does not look at forward linkages in terms of how an industry's production is used as an input for other production or for final use (or downstream effects). Hence, this study does not

[4] The IMPLAN (Impact Analysis for Planning) model is an input-output (I-O) model that measures how many times a dollar is re-spent in, or "ripples" through the economic region. The I-O model yields multipliers that are used to calculate the indirect and induced economic effects in terms of jobs, labor income, and business sales generated within a specified region per dollar of industry spending on various types of goods and services.

measure downstream effects of the Alaska pollock fishery (i.e., distribution to wholesale and retail establishments restaurants).

Employment

Table 4 shows the employment effects of the Alaska pollock industry in 2023. The industry supported an estimated 16,821 direct, indirect, and induced jobs in the U.S. The industry directly employed 8,277workers from various states nationwide—31% were Alaskans, 23% were from Washington, 3% were from Oregon, and 32% were from other states. An additional 1,089 workers were residents of other countries (12%).

Table 4. Direct, Indirect, and Induced Employment by Location

	Employment			
Location	Direct	Indirect	Induced	Total
Alaska	2,869	2,386	1,064	6,318
Washington	2,168	1,457	1,419	5,045
Oregon	265	36	121	421
AK, WA, OR Subtotal	**5,302**	**3,878**	**2,604**	**11,784**
Rest of U.S.	2,975	322	1,739	5,036
Nationwide Total	**8,277**	**4,201**	**4,343**	**16,821**
Rest of the World	1,089	0	0	1,089
Global Total	**9,366**	**4,201**	**4,343**	**17,910**

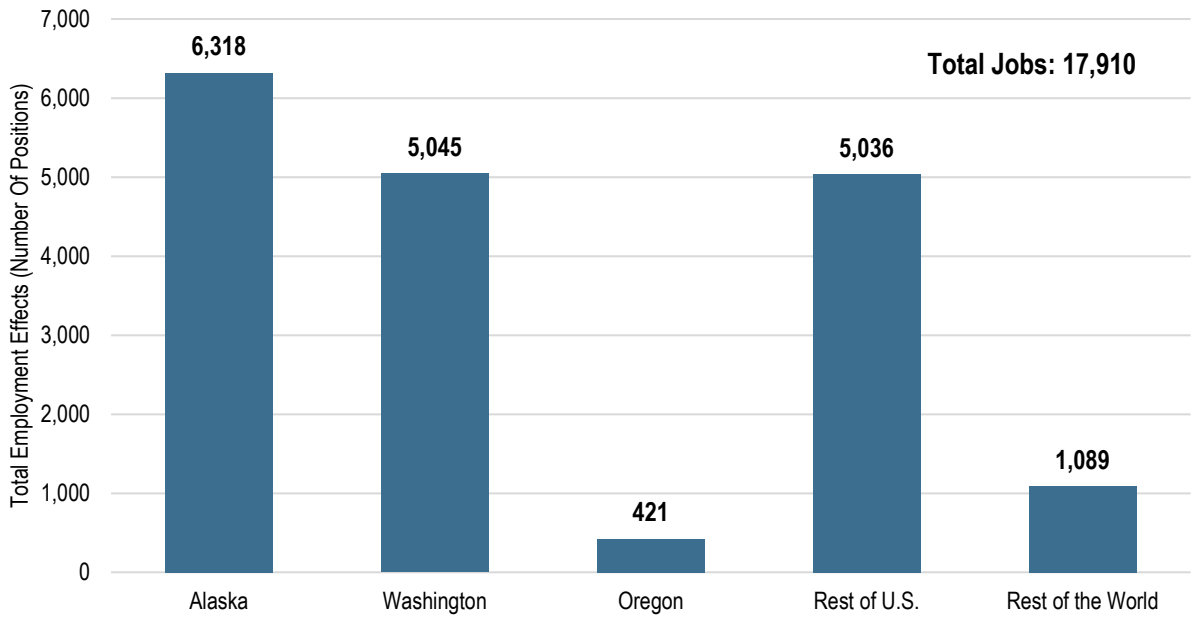
Source: Northern Economics Analysis

Direct employment is the estimated number of employees supported directly by the Alaska pollock industry. Indirect employment is the estimated number of jobs supported in sectors that provided goods and services to the Alaska pollock industry for annual operations and maintenance as well as capital spending in 2023. Induced employment includes jobs supported in sectors that provide goods and services to households, through the spending of wages of the individuals who worked in the Alaska pollock fish harvesting, processing, and seafood company sectors. These induced effects were generated in the areas where the workers live.

In addition to the direct jobs, spending by the Alaska pollock industry supported an additional 8,544 indirect and induced jobs nationwide. About 76% of these additional jobs were in Alaska (40%), Washington (34%), and Oregon (2%).

Washington had the most direct and induced employment, closely followed by Alaska. However, Alaska had more indirect jobs supported by the Alaska pollock industry than Washington. As a result, the Alaska pollock industry had the greatest employment effects on Alaska out of the three states analyzed individually at 6,318 jobs, compared to 5,045 in Washington, and 421 in Oregon (Figure 4). Additionally, about 70% of the Alaska pollock industry effects nationwide in 2023 were in Alaska, Washington, and Oregon.

Figure 4. Total Employment Effects by State or Region



Source: Northern Economics Analysis

Labor Income

Table 5 shows the labor income effects of the Alaska pollock industry in 2023. The Alaska pollock industry paid about $459 million in direct labor income nationwide; these were the payments to individuals who were directly employed by the Alaska pollock industry in 2023. The estimated additional indirect and induced labor income generated nationwide amounted to about $694 million. Combined, the Alaska pollock industry generated an estimated $1.2 billion in labor income in 2023.

The direct labor income is comprised of wages and salaries paid to the individuals directly employed by the companies that participated in the 2023 Alaska pollock fishery. Indirect income effects include wages paid for the jobs in sectors that provided goods and services to the Alaska pollock industry for annual operations and maintenance as well as capital spending (the indirect jobs supported in the *Employment* section above). Induced income effects are the wages associated with the jobs supported by the spending of the Alaska fishing industry's wages (i.e., wages of crew, processing workers, and other employees of fishing companies supported by the Alaska pollock industry).

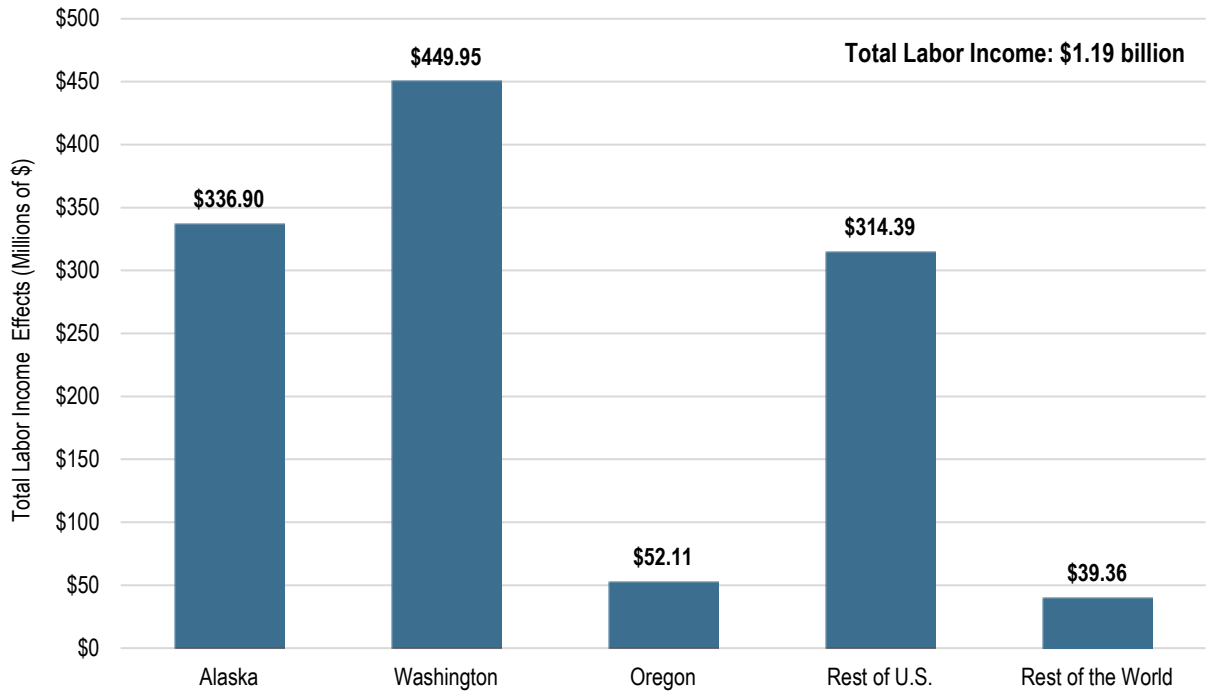
Table 5. Labor Income effects of the Alaska Pollock Industry in 2023

Location	Labor Income			
	Direct	Indirect	Induced	Total
Alaska	$102,913,225	$165,626,096	$68,360,728	$336,900,050
Washington	$171,675,711	$170,519,356	$107,756,408	$449,951,474
Oregon	$17,454,328	$22,232,360	$12,427,625	$52,114,313
AK, WA, OR Subtotal	**$292,043,264**	**$358,377,812**	**$188,544,761**	**$838,965,837**
Rest of U.S.	$167,132,990	$27,241,557	$120,017,217	$314,391,765
Nationwide Total	**$459,176,254**	**$385,619,369**	**$308,561,979**	**$1,153,357,602**
Rest of the World	$39,363,570	$0	$0	$39,363,570
Global Total	**$498,539,824**	**$385,619,369**	**$308,561,979**	**$1,192,721,171**

Source: Northern Economics Analysis

Washington experienced the largest labor income effects of the three states analyzed individually, which is interesting considering Alaska had more overall employment supported by the industry than Washington (Figure 5). One possible explanation for this could be employees earning higher wages in Washington than in Alaska, potentially due to more full-time workers in companies with headquarters/offices in Washington compared to a higher number of seasonal jobs in Alaska. Collectively Alaska, Washington, and Oregon accounted for 70% of labor income effects of the Alaska pollock industry in 2023.

Figure 5. Total Labor Income Effects by Region or State



Source: Northern Economics Analysis

Economic Output (Total Business Sales)

For this study, the economic output measures the total business sales generated by the Alaska pollock fishery in each state as a result of industry-wide spending in 2023. The estimated total business sales generated by the Alaska pollock industry in the U.S. in 2023 amounted to about $2.54 billion, with $1.1 billion in direct economic output and $1.46 billion in indirect and induced economic output. The direct economic output is the total estimated direct spending by the Alaska pollock industry for labor and non-labor operations and maintenance, plus capital spending. Indirect economic output effects are the amount of total business activity generated in various supplier sectors resulting from demand for goods and services by the Alaska pollock industry's operations in 2023. Induced economic output effects are the amount of business sales generated as a result of workers' spending of wages.

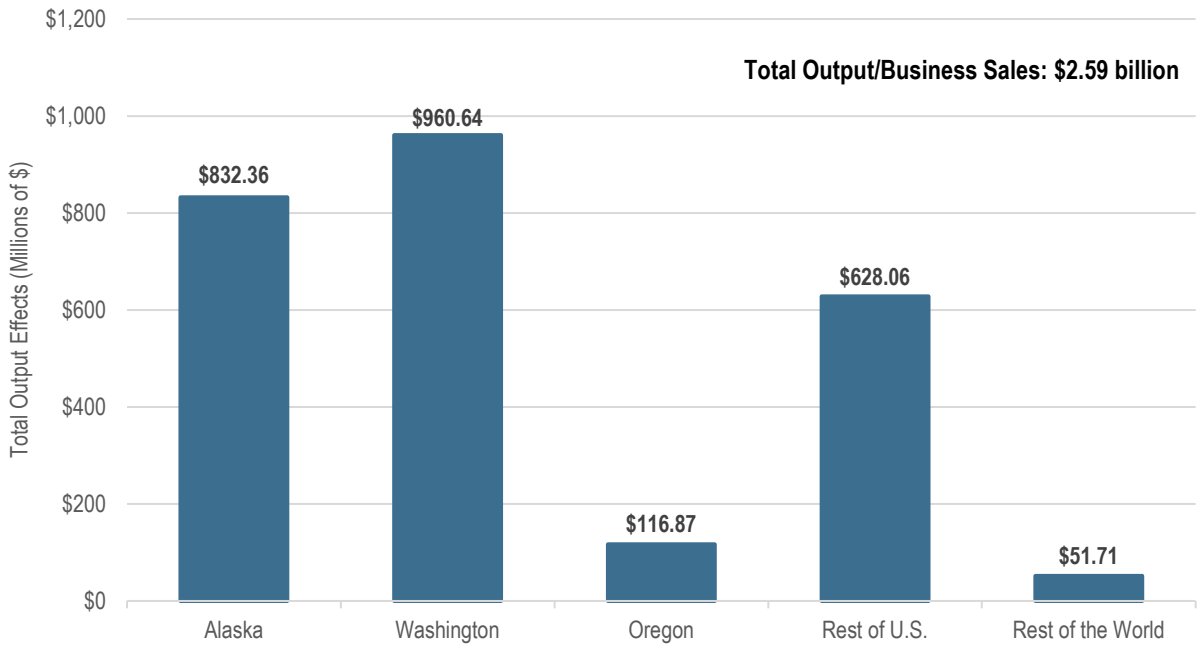
Table 6 shows the estimated economic output by location of the Alaska pollock industry in 2023. Washington experienced the largest economic output effects of the Alaska pollock industry of the three states analyzed individually, with Alaska not too far behind. Alaska, Washington, and Oregon combined accounted for about 75% of the nationwide economic output of the Alaska pollock industry in 2023 (Figure 6).

Table 6. Estimated Business Sales (Economic Output) by location of the Alaska pollock industry in 2023

	Business Sales (Economic Output)			
Location	Direct	Indirect	Induced	Total
Alaska	$410,181,534	$231,640,814	$190,542,512	$832,364,860
Washington	$407,662,842	$228,250,766	$324,724,405	$960,638,014
Oregon	$50,223,946	$31,666,238	$34,977,222	$116,867,406
AK, WA, OR Subtotal	**$868,068,322**	**$491,557,818**	**$550,244,140**	**$1,909,870,280**
Rest of U.S.	$207,605,675	$47,220,244	$373,232,476	$628,058,396
Nationwide Total	**$1,075,673,998**	**$538,778,062**	**$923,476,616**	**$2,537,928,676**
Rest of the World	$51,712,603	$0	$0	$51,712,603
Global Total	**$1,127,386,600**	**$538,778,062**	**$923,476,616**	**$2,589,641,279**

Source: Northern Economics Analysis

Figure 6. Total Output Effects by Region or State



Source: Northern Economics Analysis

Fiscal Impacts

There are a number of taxes that vessels and shorebased processors must account for during operation that are applied by operation type and calculated using reported revenue. Taxes on fisheries in Alaska include the Fisheries Business tax, which is shared with the local borough, the Fishery Landing Tax, the Seafood Marketing Assessment, and any applicable local taxes of landed seafood products. The Fisheries Business Tax is levied on both floating processors, such as motherships and catcher-processors, and shorebased processors.

Taxes paid to the state of Alaska, local boroughs, and cities can be estimated based on reported landed volumes and values of Alaska pollock in 2023 (NOAA Fisheries 2024) (Table 7ES Table). Across state fisheries business taxes, fishery resource landing taxes, any local or borough raw fish taxes, and other taxes (such as the Seafood Marketing Assessment tax), it is estimated that the Alaska pollock industry generated $21 million in Alaska local, borough, and state tax revenue in 2023.

Table 7. Estimated 2023 Alaska Pollock Fisheries Taxes

Tax Type	2023 Estimate
State Fisheries Business Tax and Fishery Resource Landing Tax	$13,846,145
Local Raw Fish Taxes	$4,997,471
Other	$2,307,691
Total	**$21,151,307**

Note: Only Alaska state fishery taxes are estimated. Other taxes include seafood marketing assessment taxes.

Source: NOAA Fisheries 2024 and Northern Economics analysis

Summary of Effects on Alaska

Table 8 below summarizes the economic and fiscal effects of the fishery in the state of Alaska in 2023. This includes impacts stemming from the purchases of fuel, food, and other inputs for harvesting and processing operations; services utilized by fishing vessels while in Alaska ports; and any other equipment or machinery purchased in the state. It also includes labor income earned by Alaskans employed on vessels, who work in processing plants, or who are employed by companies or CDQ groups in the state.

In 2023, an estimated 2,869 Alaskans were employed by the Alaska pollock fishing industry. This includes fishing vessel captains and crew, processing and non-processing crew workers, and shorebased processing workers. It also includes those who do not necessarily work on vessels or at seafood processors but who work for seafood companies, companies that own vessels, or CDQ groups, whose positions are supported by the revenue generated from the operation of fishing vessels or processors and/or Alaska pollock quota royalties. Furthermore, 2,386 Alaska jobs were indirectly supported in sectors that provided goods and services related to the operations of Alaska pollock fishing vessels, catcher-processors, shorebased processors, CDQs, and companies, and 1,064 additional Alaska jobs (induced jobs) were supported in sectors that provided goods and services to Alaska-based employees (through spending of their wages). Combined, the fishery supported a total of 6,318 jobs in the state in 2023.

Alaskan industry workers earned nearly $103 million in wages and benefits in 2023, and similar to employment effects, the purchasing of goods and services generated $165.6 million in indirect labor income and $68.4 million in induced income.

In total, direct economic output, including all direct labor income and the purchases of goods and services in Alaska in 2023 is estimated to be $410 million, and when including indirect and induced output effects, the economic contribution of the industry more than doubles to $832 million.

In addition to these employment and spending-related effects, we also estimate the fiscal impacts of the Alaska pollock industry in the state. In 2023, we estimate that the fishery generated an additional $21 million in Alaska local, borough, and state tax revenue.

Table 8. Estimated Economic Effects of the Alaska Pollock Fishery in Alaska

Economic Effect	Direct	Indirect	Induced	Total
Employment	2,869	2,386	1,064	6,318
Labor Income	$102,913,225	$165,626,096	$68,360,728	$336,900,050
Economic Output/Business Sales	$410,181,534	$231,640,814	$190,542,512	$832,364,860

Source: Northern Economics Analysis

It is important to note that the estimates here include the direct, indirect, and induced effects of the CDQ groups' operational expenses and employees, but do not include the indirect and induced effects associated with the millions of dollars spent on programs and projects in the state. Because of this, the estimated total economic contribution of the Alaska pollock Fishery is underestimated. As described in the CDQ annual reports, a significant proportion of CDQ spending is targeted to supporting local fishing industries and communities, through grants and scholarships, training programs, and other purposes. In total, direct spending on programs and projects is estimated to total over $51.7 million in 2023, approximately 32% of total spending, based on interviews and information available in annual reports. Due to the vast number of these programs and the uniqueness of each CDQ, much more information is needed to understand specifically what types of spin-off (indirect and induced) economic effects such programs have on the Alaska state economy and therefore cannot be estimated here. Quantifying the spin-off effects of their programs and special projects would require a more detailed accounting of their spending on goods and services and where the spending occurred for each program which is outside the scope and budget for this study.

Conclusions

In 2023, the Alaska pollock fishery is estimated to have supported over 16,821 jobs across the U.S., with the majority in Alaska (6,318) and Washington (5,045). These jobs in turn support economic activity through wages and the spending of wages, particularly in Alaska and Washington, with $337 million in estimated total labor income effects in Alaska and $450 million in Washington. Across the U.S., labor income effects alone are estimated to be over $1.5 billion.

When combined with the impacts generated by the purchasing of supplies, services, and equipment by Alaska pollock fishing vessels, processors, companies, and CDQ groups, the total economic output effect in the US is estimated to be $2.54 billion, with nearly a billion dollars in effects in Washington ($961 million) and $832 million in impacts in Alaska alone. In Alaska, impacts are even greater when considering the fiscal impacts ($21 million in state, borough, and local taxes). Furthermore, this study does not include indirect and induced effects of CDQ spending on special projects and programs, indicating the total estimated effects on Alaska are likely underestimated.

Methods

Survey

Survey Design & Testing

To estimate indirect and induced effects of the Alaska pollock fishery, estimates of total direct employment and spending were needed. Surveys of Alaska pollock fishery businesses were created to gather representative information from all relevant sectors on their employment and spending patterns (*Appendix: Survey Instruments*). In total, there were five surveys: catcher-vessels, motherships, catcher-processors, shorebased processors, and one for other employment and spending not captured in individual vessel surveys (e.g., of headquarters or corporate offices). Survey questions were designed to solicit information on total employment and spending by the vessel, processor, or company that is directly attributable to the Alaska pollock fishery as well as the location where employees reside or where spending occurred. For some spending categories where spending was not expected to be specific to the Alaska pollock fishery (e.g., some capital costs), questions solicited information on total revenue or effort (days at sea) across various activities to facilitate cost disaggregation.

Draft survey instruments were sent to APFA first for their review and feedback based on their knowledge of the fisheries. Then, knowledgeable representatives from each sector were contacted to review the survey instrument to ensure the survey questions were clear and worded appropriately for the sector. Once edits and revisions to the survey instruments were made, a revised version of the survey was provided for final review and distribution. To facilitate completion of the survey for companies that own multiple vessels and/or processors, excel versions of the survey were also provided, though the primary version of the survey was provided as a fillable PDF.

Survey Distribution

APFA distributed surveys to representatives across sectors and regions alongside survey instructions as well as non-disclosure forms. Signed non-disclosure agreements were collected by APFA while survey results were collected by Northern Economics alone. The survey was first distributed on January 16th, 2025, and was open until March 12th, 2025. During this time, Northern Economics staff reviewed individual survey responses as they came in and conducted follow-up with respondents to revise or clarify responses as needed (additional information provided in the next subsection).

Survey Analysis

The number of vessels, processors, and companies captured in the survey is summarized below in Table 9. This does not capture the individual number of respondents, since in many cases, multiple

vessels or processors were owned by the same individual or company and multiple survey responses were submitted by one entity.

Table 9. Survey Responses by Sector

Sector/Type of Survey	Number of Surveys
Catcher Vessels	33
Motherships	3
Catcher-Processors	10
Shorebased Processors	6
Other Employment and Spending	12

Source: Northern Economics Analysis

As survey responses were received, each response was reviewed by a project team member for potential errors, including duplications, omissions, anomalous values, or other issues. A second team member would then review the survey responses against the list of potential errors and questions for follow-up with the respondent and modify as necessary before reaching out to the respondent.

Once all responses had been reviewed and cleaned, analysis of the survey data was initiated. Data were first disaggregated to amounts directly attributable to the Alaska pollock fishery. Days at sea spent across various Alaska pollock and non-pollock fisheries were used for disaggregation when the amount directly attributable to Alaska pollock was unable to be determined (e.g., in the case of capital expenditures). Then, amounts were further distributed to the location of spending or in the case of employment, residency, based on responses.

Based on the reviewed and cleaned data, any obvious outliers and incomplete responses were removed from their respective datasets. Obvious outliers were those that had atypical operations for their sector to the best of our team's knowledge and the other data provided. One catcher vessel individual survey response was removed from the baseline estimates used for survey extrapolation because it was incomplete and was anomalous compared to other vessels, and our subsequent follow-up efforts to clarify and complete responses were unsuccessful. Additionally, two other responses were included in the baseline assessment of impacts (direct impacts) but were not used to develop average estimates used for extrapolation purposes. Specifically, we removed one shorebased processor response from the analysis since it was not representative of the sector more broadly, due to the nature of processing operations. One catcher processor response was also removed because it was a new vessel, so the survey response mostly contained "NA" or blanks for the questions since it did not actually fish in the survey year. For both of these cases, reported employment and spending was captured in the analysis under the estimate of direct effects, but did not contribute the estimate used to extrapolate spending and employment patterns for unobserved vessels.

Extrapolation

In total, in comparison to reported survey harvest or purchased volumes against AFA cooperative reports[5] and NOAA Fisheries Groundfish Economic SAFE[6] reported harvest volumes (NOAA FISHERIES 2024), the survey results captured approximately 34% of catcher vessel harvest, 77% of catcher-processor harvest, 100% of mothership purchases, and 60% of shorebased processor purchases of Alaska pollock in 2023. In total, for the companies that submitted a survey of other employment and spending, this covered 36% of all harvest or purchase volume, noting that it is possible that not all vessels or processors are part of larger companies and may not have had other employment or spending to report (see *Company Data Extrapolation* for more information). Table 10 shows survey response coverage of population by sector in more detail.

Table 10. Survey Response Coverage of Population by Sector

Sector	NOAA Reported Harvest or Purchase Volumes (mt)	Survey Reported Harvest Volumes (mt)	Estimated Survey Coverage (%)
Catcher Vessel	810,670	273,153	34
Catcher-Processor	451,685	347,604	77
Mothership	112,793	112,793	100
Shorebased Processor	697,877	420,705	60
Company	2,073,025	753,231	36

Source: NOAA, NOAA FISHERIES, and Northern Economics analysis

If the survey sampled less than 100% of harvest for a given sector, survey responses were used to extrapolate results for the remaining harvest volume in the sector (using data from NOAA Fisheries (NOAA Fisheries 2024) and AFA cooperative reports. Extrapolations were based on average values per metric ton (landed or purchased weight) calculated for each variable within each sector after removing any outliers (identified generally as more than 1.5 times the interquartile range of the variable). This extrapolation procedure was selected as the most suitable based on the limited number of observations, limited information about the population of GOA catcher vessels, and evidence of linear relationships between study variables of interest (principally input costs) and landed/purchased weight. For harvesting sectors, relationships between days at sea were also explored but not selected due to weaker correlations. Landing or purchased weight was the most readily observable measure of the unsampled proportion of the entire Alaska pollock fishery for each sector for the study year.

Average input cost estimates used for extrapolation were derived differently for capital costs than other operating and maintenance costs. For capital cost estimates, the survey team identified and removed survey points that were over $400,000 for any given capital cost category. This was because

[5] *https://www.npfmc.org/cooperative-reporting/*

[6] *https://www.fisheries.noaa.gov/alaska/ecosystems/economic-status-reports-gulf-alaska-and-bering-sea-aleutian-islands*

it appeared that either respondents reported non-capital spending (unusually low or costs that appeared to scale with effort/landings) and that after removing high values, calculating an average spend per metric ton appeared justified. It may be that firms either reported O&M spending in the capital cost section or that accounting procedures and practices may vary across firms and firms reported their capital spending in different ways. It should be noted that any capital costs reported in the survey were included in the analysis, including the purchase or construction of new vessels, but these capital costs were not extrapolated to the unobserved portion of the fleet. This is because the survey was not extensive enough to understand at what point any given operation is in its business cycle and cannot predict inherently lumpy (infrequent, large) expenditures across the rest of the fleet. In this respect, the analysis is conservative in its estimation of industry-wide expenditures.

CDQ Interviews

Estimates of spending and employment associated with the Alaska pollock fishery were generated differently for Western Alaska CDQ groups. Interviews were conducted with representatives from the CDQ groups to verify published information from their 2023 annual reports on their operations, employment, and spending. In addition, these interviews were used to obtain an estimate of the proportion of their total revenue attributable to the Alaska pollock fishery in 2023, specifically revenue stemming from the ownership of fishing vessels or processors or from quota royalties. The level of information requested was similar to the level requested in the survey of other employment or spending (company survey) and a spreadsheet for review was pre-filled before the interview with information from their 2023 annual CDQ report.

CDQs were asked about total employees, their compensation (including bonuses and benefits, taxes, and unemployment), general and administration expenses, programs and project spending, and fishery operating and capital expenditures not on Alaska pollock vessels (since that would already be captured elsewhere in the study). For all employment and spending, the number of employees who are Alaska residents or percent of spending occurring in Alaska was also requested. Only one of the six CDQ groups required assumptions about information that was not provided in its 2023 annual report. Here, averages from other participating CDQs were applied and adjusted by the relative size of total revenue, employment, or spending. In addition, only the proportion of spending inside of Alaska was requested to simplify the information needed for the study. Overall, respondents indicated that the majority of spending and employment occurs in Alaska. As noted previously, while the interviews asked for verification about the total amount of projects or program spending, due to the increased level of information needed to understand how various programs and projects stimulate spending and/or employment throughout the state, multiplier (indirect and induced) impacts for this spending are not included in results.

I-O Analysis

This study quantified the economic contribution of the Alaska pollock Fishery to Alaska (AK), Washington (WA), Oregon (OR), and the rest of the United States (RUS). The economic contribution was measured in terms of jobs, income, and business sales (or economic output). The Alaska pollock fishery (Industry), as defined in this study, includes the catcher vessels, catcher processors, motherships, shore-based processors, companies, and CDQ groups that fished and/or processed Alaska pollock, owned and operated vessels or processors, or in the case of CDQ groups, received quota royalties in 2023. Note that the analysis of the CDQ groups was focused only on their economic contribution to Alaska as there was not enough information on their spending and employment in other states.

Input-output (I-O) analysis is a typical approach used to measure the effects of an activity or evaluate the economic contribution of an entity or an industry to a specified region, in this case AK, WA, OR, and RUS. I-O analysis yields estimates of the direct, indirect, and induced effects of industry spending on goods and services, in this case the Alaska pollock fishery industry spending on annual operations and capital projects in 2023.

The analysis is based on a model of the inter-industry transactions within each of these regions. The I-O model is a matrix that tracks the dollar flow between the industries within the economic region of interest. The model can measure how many times a dollar is re-spent in, or "ripples" through, a community (or a larger economic region) before it leaks out. The I-O model yields multipliers that are used to calculate the indirect and induced effects on jobs, income, and business sales/output generated per dollar of spending on various types of goods and services in the study area. To evaluate the economic effects to the state or a particular region, only the "local" (i.e., within the state or within the region) expenditures are used in the model; the rest are considered leakages. More leakages mean smaller multipliers, and the larger the local expenditures, the greater the multiplier effects. The multipliers for any given industry in any given location are unique, based on industry composition and geographic area.

In I-O models, multipliers measure only the backward linkages, also known as upstream effects. Input-Output analysis does not look at forward linkages in terms of how an Industry's production is used as an input for other production or for final use, also known as downstream effects. This study therefore does not consider the downstream effects of the Alaska pollock fishery (i.e., employment, income, and business sales associated with the downstream distribution of Alaska pollock to wholesale, retail, and dining establishments.

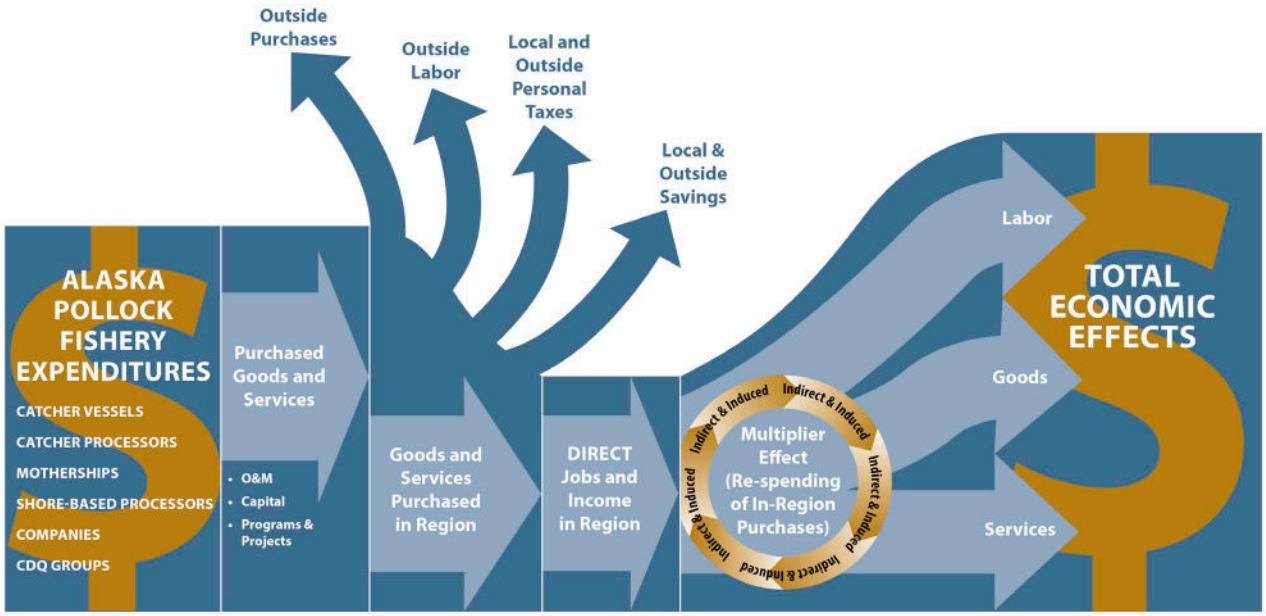
IMPLAN software and data were used to develop the I-O model for the various regions. IMPLAN is a software package widely used for economic impact assessments. It uses specific data on what inputs are needed in the production of goods or provision of services for over 500 economic sectors. IMPLAN also has area-specific data on what industries are available locally from which to purchase

those inputs. The study applied the most recent (2023) IMPLAN data on multipliers for all the economic sectors in the I-O model.

The data collected on in-state spending (labeled as "project expenditures" in the figure) by the industry are the inputs to the IMPLAN model. An extensive survey effort was conducted from January 2025 to March 2025 to collect direct employment and expenditure data from the industry (see the *Survey Methodology* section above). The direct expenditures and wages fuel the economy as a portion of these amounts is spent at other businesses in the region, around the state, and outside the state. The recipients of these initial expenditures repeat the process of paying wages and spending on goods and services, until all the money has leaked from the local and state economies into outside economies and savings. This process is known as the multiplier effect and is measured as the number of times a dollar of direct spending is re-spent in the region before it leaks out. The cumulative sum of the original jobs and expenditures and the indirect and induced jobs and expenditures created by the money flowing through the economy constitutes the total economic effects.

Figure 7 demonstrates the framework for estimating the economic effects using input-output analysis.

Figure 7. Framework for Evaluating Total Economic Effects of Industry Spending



Source: Northern Economics, Inc.

The I-O Process

The following provides a general outline of the steps applied in the economic modeling for this study:

Step 1: Compiled spending data by category for all Industry sectors.

The total annual spending was broken out into the categories shown in Table 11, which defines annual operating expenses and capital spending categories for catcher vessels (CV), catcher processors (CP), and motherships (MS), and Table 12, which defines those categories for shore-based processors.

Table 11. Annual Operating Expenses and Capital Spending Categories for Catcher Vessels (CV), Catcher Processors (CP), and Motherships (MS)

Category	Item
Labor Expenses (Operating Expenses)	Total Labor Payments (CV)
	Payments to Processing and Non-Processing Crew (CP, MS)
Non-Labor Expenses (Operating Expenses)	Food
	Fuel and lubrication
	Bering Sea Pollock observer fees, electronic monitoring, and reporting costs
	Satellite/VMS or other communication
	Freight
	Travel
	Offloading (cross-dock fees, tariffs, hoist fees, etc.)
	Fishing association fees
	Cooperative fees
	State and federal permit/license fees
	Lease expenses for vessels and onboard equipment
	Repair and maintenance expenditures (e.g., nets, doors, cables, etc. fully expensed in 2023)
	Insurance associated with vessel operations (excluding employee health insurance)
	Moorage and wharfage
	Management fees (paid to management companies)
Capital Expenses	Vessel and onboard equipment
	Fishing gear (CV and CP only)
	Processing equipment (CP and MS only)
	Capitalized repair and maintenance (CV only)

Table 12. Annual Operating Expenses and Capital Spending Categories for Shore-based Processors

Category	Item
Labor Expenses (Operating Expenses)	Production workers' wages
	Other workers' wages
Non-Labor Expenses (Operating Expenses)	Rent or lease of buildings or other structures
	Rent or lease of processing machinery or equipment
	Maintenance and repair of buildings, machinery, or equipment
	Utilities (e.g., electricity, gas, water, sewer)
	Monitoring costs (shoreside only)
	Offload expenses paid to other facilities
	Insurance
	State and federal permit/license fees
	Licensing fees
	Communications and office supplies
	Freight costs of supplies
	Production supplies (e.g., knives, hairnets)
	Other supplies (cleaning, additives, etc.)
	Other Misc. Overhead (Employee Travel, Fisherman Relations)
Capital Expenses	Buildings
	Machinery and equipment

As noted in *CDQ Interviews* above, information on annual general and administrative expenditures, programs and special projects, and non-pollock fishery operating and capital spending by the CDQ groups was obtained from their 2023 annual reports and interviews.

There were a few interim steps and adjustments made to the data in the modeling process, including, among others, the following:

- A process called "margining" or adjusting the spending on commodities or goods to only apply the transportation, retail, and wholesale trade margins to the model. These are supplies that were purchased locally (from wholesale or retail establishments), but that were not manufactured in the region.
- Assigning appropriate multipliers and determining share of spending for categories that encompass more than one sector (i.e., travel expenses were split among accommodations (80%), restaurants (10%), and transportation (10%); repair and maintenance expenses were split between machinery and equipment repair (50%) and maintenance sector, and support activities for transportation sector (50%); spending on utilities was assigned to electric power generation systems, electric power transmission and distribution, and water, sewage and other systems, depending on what are available in each of the regions).

Step 2: Allocated the spending amounts under each category to the region/state where the spending occurs.

The survey asked respondents to indicate the amount/percent of expenditure spent within Alaska, Washington, Oregon, rest of the US, and rest of the world (ROW). For the CDQ groups, only the amount spent in the state of Alaska was available.

Step 3: Assigned the non-labor O&M spending amounts in each region to the appropriate IMPLAN model economic sectors.

Since the multiplier effects are unique for each sector and vary by region/state, the spending is matched to a specific IMPLAN sector in the state I-O model, as shown in Table 13 below, and the model applies the appropriate multipliers for that sector or combination of sectors. This is referred to as "analysis by parts" since there is no one sector in the model that can specifically approximate the spending patterns of the Alaska pollock fishery sectors. These sector assignments were similar to the sector assignments used in other fisheries I-O models including the Input-Output Model for the Pacific Coast Fisheries (IO-PAC); Alaska head and gut fishing fleet multiregional social accounting matrix (MRSAM) model for Alaska, West Coast, and Rest of the USA; the Northeast Region Commercial Fishing I-O model (NERIOM); and NOAA Fisheries Commercial Fishing & Seafood Industry Input/Output Model (CFSI I/O Model).

Table 13. Allocation of Non-Labor O&M Spending by Economic Sector

Category	Sector #	Sector Description
Food	381	Wholesale - Grocery and related product wholesalers
Fuel and Lubrication	3146	Refined petroleum products (commodity)
Observer fees, electronic monitoring, and reporting costs	446	Scientific research and development services
Satellite/VMS or other communication	417	Satellite, telecommunications resellers, and all other telecommunications
Freight	402	Support activities for transportation
Travel	489	Hotels and motels, including casino hotels
	491	Full-service restaurants
	400	Transit and ground passenger transportation
Offloading (cross-dock fees, tariffs, hoist fees, etc.)	402	Support activities for transportation
Fishing association fees	505	Business and professional associations
Cooperative fees	506	Labor and civic organizations
State and federal permit/license fees	Federal Gov't	Institutional Spending Pattern
	State/Local Gov't	Institutional Spending Pattern
Lease expenses for vessels and onboard equipment	435	Commercial and industrial machinery and equipment rental and leasing
Repair and maintenance expenditures	497	Commercial and industrial machinery and equipment repair and maintenance
	402	Support activities for transportation
Insurance associated with the operation of this vessel	426	Insurance carriers, except direct life
Moorage and wharfage	402	Support activities for transportation
Management fees (paid to management companies)	451	Management of companies and enterprises

Step 4: Adjust and assign spending on wages and salaries to the appropriate regions.

For labor expenses (wages and salaries), the amounts from the survey are adjusted to generate estimates of personal disposable income (adjusting for taxes and savings based on average rates specific to the household income levels) and the amounts are allocated to the regions where the employees reside. The economic/stimulus effects of spending of income occur where the employees reside and spend their income, which is not necessarily the same as where they work or where the harvesting or processing activity occurred.

Step 5: Run the model and generate employment, income, and economic output/business sales effects.

The IMPLAN model outputs are incorporated into a spreadsheet model developed in Excel. This spreadsheet model generates a summary of the direct, indirect, and induced effects by spending category and by region.

Fiscal Analysis

Alaska Fishery Resource Taxes

All Alaska pollock harvesting and processing sectors are affected by various seafood and fishery taxes based on the location of harvesting or processing activities. In this report we estimate taxes paid to the state, boroughs, or local communities within Alaska based on the reported average ex-vessel prices and harvest volumes for each sector and region in 2023 (NOAA Fisheries 2024). A table of the tax rates by sector is shown in Table 14. The Alaska State Fisheries Business Tax is levied on persons who process or export fisheries resources from Alaska and is based on the ex-vessel value of the harvest. The tax rate varies by sector with motherships and catcher-processors assessed at 5% of ex-vessel value and shorebased processors assessed at 3% of ex-vessel value (Spanos 2019). Any other floating fisheries business that does not fall under the Fisheries Business Tax would need to pay the Fishery Landing Tax, which applies to floating processing activities in federal waters who bring their products into Alaska for transshipment. This tax is assessed at 3% of ex-vessel value for established species like Alaska pollock. Both taxes are shared with municipalities and boroughs, if applicable. The final Alaska state tax that is assessed at 0.5% of revenue is the Seafood Marketing Assessment for any entity that processed, landed, or exported $50,000 or more in a calendar year (Spanos 2019). For the purposes of our analysis, we assume all harvest of Alaska pollock by shorebased processors paid the Alaska State Fisheries Business Tax, and all at-sea processing sectors paid the Fishery Landing tax, since it is not known what proportion of at-sea processors land their products in Alaska for transshipment, though it is likely less than 100%, meaning this is likely an overestimate of the total tax paid.

Local taxes are only for the revenue from the harvest landed in that borough or city. In this analysis, we assumed that BSAI CV harvests were landed in the City of Unalaska, Sand Point, or King Cove, which all have a local raw fish tax of 2%. For shorebased processors in the Gulf of Alaska we assume landing occurred in all the BSAI communities as well as Kodiak which has a Sea Product Severance Tax of 0.925%. Because it was not known what proportion of GOA harvests were landed across these communities in 2023, the assumed raw fish tax rate of 2% was applied, which likely overestimates taxes paid to Kodiak. If all GOA CV harvests were subject to the Kodiak raw fish tax rate of 0.925%, this would reduce our estimate of local taxes by $408,008 or 8% of the total estimated local tax revenue (Table 7).

Table 14. Alaska State and Local Fisheries Tax Rates

Tax Type	Shorebased Processor	Motherships	Catcher-Processors
Fisheries Business Tax[1]	3% for established species	5% for established species	5% for established species
Fishery Landing Tax[1]	-	3% for established species	3% for established species
Seafood Marketing Assessment[1]	0.5%	0.5%	0.5%
City of Unalaska Raw Seafood Tax[2]	2%	-	-
Kodiak Island Borough Sea Product Severance Tax[3]	0.925%	-	-
Sand Point Raw Fish Tax	2%	-	-
King Cove Fish Processing Business Tax	2%	-	-

Sources: [1] *Spanos, Brandon (2019) "Fish Tax Overview" presented to the Houser Resources Committee April 24, 1019.* https://www.akleg.gov/basis/get_documents.asp?session=31&docid=35723

[2] *City of Unalaska;* https://www.ci.unalaska.ak.us/cityclerk/page/raw-seafood-tax

[3] *Kodiak Island Borough;* https://www.kodiakak.us/284/Severance-Taxes

Key Assumptions

Accuracy of Survey and Interview Responses

Our analysis assumes that survey and CDQ interview respondents understood the questions and provided accurate estimates of their spending and employment activities. While the survey team reviewed each response for potential issues and asked respondents for clarification or revision when necessary and removed outliers from the dataset, the analysis is particularly reliant on respondents to provide accurate information about where workers reside and to provide the location where spending occurred. In the survey, respondents were instructed to provide their best estimate of where spending occurred—specifically, "Assign the location based on your **best estimate** of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice." In interviews, participants were instructed to indicate the proportion of spending that occurred in Alaska. The analysis therefore relies on these good-faith estimates of where spending occurred and uses the average across respondents to estimate spending amounts and the proportion of spending by location for the unobserved proportions of each sector.

Extrapolation of Data

For sectors where extrapolation was necessary, the analysis assumes that survey responses are from a representative sample of the sector. The extrapolation analysis further assumes that most spending is proportionate to effort in the fishery and uses average cost and employment estimates per metric ton to estimate unobserved values. For most operating costs, this assumption is warranted; however, for some of the capital cost categories, this assumption does not hold. For new vessel purchases and capital spending on building expansions for example, these capital costs by nature do not scale with effort and are relatively unpredictable. Because of this, our analysis does not attempt to estimate these capital expenditures for the unobserved portions of each sector.

The extrapolation analysis further assumes that for employment and spending not directly associated with harvesting and processing activities (reported on the survey as 'other employment and spending'), companies that submitted survey responses for a vessel or processor but did not submit the 'other employment and spending' survey did not have any additional spending or employment to report. However, while this is likely true for many survey respondents, it may also be that some survey respondents just decided not to fill out that survey. Because of that, we also explored a conservatively high scenario where we assume that all companies have company spending similar to the respondents of that survey. The results of this sensitivity analysis are reported in the next section.

For CDQs, while only 1 CDQ group required any extrapolation, the effects of this extrapolation are assumed to be minimal, because most information on employment and spending for the CDQ groups was available in their annual reports. Gaps that needed to be filled included the proportion of revenue attributable to Alaska pollock, and percent of spending in Alaska. For both assumptions, limited variability across CDQ groups who participated in the study confers confidence in the estimation of these missing pieces.

Sensitivity Analyses

To explore the robustness of our results to key assumptions, we conducted two specific sensitivity analyses: one for catcher vessel type and one for the extrapolation of other employment and spending results for company spending.

Catcher Vessel Primary Fishery Region

The first explored the assumption that catcher vessel spending (both by amount and by allocation to state) was similar across the BSAI and the GOA. In the sensitivity analysis, we estimated a BSAI and GOA catcher vessel estimate separately, based on the fishery where the vessel spent the majority of effort, and extrapolated based on the remaining volume of Alaska pollock landings in each fishery. We find that overall, impact estimates are very similar, with less than half of a percent differences between the base case and the sensitivity case.

Table 15. Catcher Vessel Fishing Region Sensitivity Results

Region	Employment		Labor Income		Output	
	Direct	Total	Direct	Total	Direct	Total
Base Scenario						
Alaska	2,869	6,318	$102,913,225	$336,900,050	$410,181,534	$832,364,860
Washington	2,168	5,045	$171,675,711	$449,951,474	$407,662,842	$960,638,014
Oregon	265	421	$17,454,328	$52,114,313	$50,223,946	$116,867,406
Rest of U.S.	2,975	5,036	$167,132,990	$314,391,765	$207,605,675	$628,058,396
Nationwide Total	8,277	16,821	$459,176,254	$1,153,357,602	$1,075,673,998	$2,537,928,676
Rest of the World	1,089	1,089	$39,363,570	$39,363,570	$51,712,603	$51,712,603
Total	**9,366**	**17,910**	**$498,539,824**	**$1,192,721,171**	**$1,127,386,600**	**$2,589,641,279**
Sensitivity Case						
Alaska	2,864	6,313	$102,245,611	$336,224,464	$409,478,448	$830,995,966
Washington	2,171	5,050	$172,459,056	$450,871,198	$408,378,299	$962,120,243
Oregon	264	414	$17,324,109	$51,503,316	$49,780,602	$116,041,612
Rest of U.S.	2,974	5,031	$167,156,869	$314,108,975	$207,347,019	$627,475,099
Nationwide Total	8,273	16,807	$459,185,645	$1,152,707,952	$1,074,984,368	$2,536,632,919
Rest of the World	1,089	1,089	$39,354,179	$39,354,179	$51,833,775	$51,833,775
Total	**9,362**	**17,896**	**$498,539,824**	**$1,192,062,131**	**$1,126,818,143**	**$2,588,466,694**
Percent Difference	**-0.04%**	**-0.08%**	**0.00%**	**-0.06%**	**-0.05%**	**-0.05%**
Difference	**-4**	**-14**	**$0**	**-$659,040**	**-$568,457**	**-$1,174,584**

Company Data Extrapolation

As described in the 'Key Assumptions' section previously, it is assumed that companies that submitted a survey for their harvesting and/or processing operations but did not submit a survey of 'other employment and spending' did not have any spending to report (the 'Base Scenario'). However, the other possibility is that they did have spending but did not submit a survey. Because of this, we constructed another scenario ('High Scenario', in Table 16 below) that assumes that all companies have similar levels of other employment and company spending as reported by those who did submit survey responses, to see how much this assumption would affect our results. We find that under the high scenario, the estimate of total employment effects is about 1% higher than under the base scenario (93 jobs), and total output effects are 1.3% higher than the base scenario, or $32.5 million. Because it is unlikely that all Alaska pollock harvesters and processors have additional non-harvesting or processing employment spending, it is likely that the actual difference between the base estimate and the true impacts are smaller than the estimated scenario differences. As a result, we conclude that the impact of our assumption has minimal impacts on our estimates. Total employment impacts are likely between 17,910 and 18,105 jobs and total output impacts are between $2.59 billion and $2.62 billion.

Table 16. Company Extrapolation Sensitivity Analysis Results

Region	Employment		Labor Income		Output	
	Direct	Total	Direct	Total	Direct	Total
Base Scenario						
Alaska	2,869	6,318	$102,913,225	$336,900,050	$410,181,534	$832,364,860
Washington	2,168	5,045	$171,675,711	$449,951,474	$407,662,842	$960,638,014
Oregon	265	421	$17,454,328	$52,114,313	$50,223,946	$116,867,406
Rest of U.S.	2,975	5,036	$167,132,990	$314,391,765	$207,605,675	$628,058,396
Nationwide Total	8,277	16,821	$459,176,254	$1,153,357,602	$1,075,673,998	$2,537,928,676
Rest of the World	1,089	1,089	$39,363,570	$39,363,570	$51,712,603	$51,712,603
Total	**9,366**	**17,910**	**$498,539,824**	**$1,192,721,171**	**$1,127,386,600**	**$2,589,641,279**
High Scenario						
Alaska	2,880	6,361	$103,347,191	$339,804,849	$414,226,900	$839,772,601
Washington	2,236	5,180	$178,366,605	$462,701,254	$418,000,543	$984,555,959
Oregon	273	430	$17,546,700	$52,242,805	$50,316,318	$117,060,698
Rest of U.S.	2,981	5,045	$167,357,497	$314,868,567	$207,930,413	$629,046,909
Nationwide Total	8,370	17,016	$466,617,993	$1,169,617,474	$1,090,474,174	$2,570,436,168
Rest of the World	1,089	1,089	$39,363,570	$39,363,570	$51,712,603	$51,712,603
Total	**9,459**	**18,105**	**$505,981,563**	**$1,208,981,044**	**$1,142,186,776**	**$2,622,148,770**
Percent Difference	**0.99%**	**1.09%**	**1.49%**	**1.36%**	**1.31%**	**1.26%**
Difference	**93**	**196**	**$7,441,739**	**$16,259,873**	**$14,800,176**	**$32,507,492**

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Appendix: Survey Instruments

This appendix includes copies of all final survey instruments:

1. Survey Instructions
2. Catcher-Processor Survey
3. Catcher Vessel Survey
4. Mothership Survey
5. Shorebased Processors Survey
6. Other Employment and Spending Survey

Survey Instructions

Northern Economics, Inc. has been contracted by the Alaska Pollock Fishery Alliance (APFA) to conduct an economic impact study of the Alaska pollock industry. To accomplish this, **five** survey instruments have been designed for individual catcher vessels, motherships, catcher-processors, shorebased processors, and parent companies to provide information on their economic activities, including employment and spending patterns. In addition, we have provided excel versions of four of these surveys that companies that own multiple vessels or shorebased processors may fill out for their fleet. Please read these instructions carefully.

Step 1: Fill out survey for individual vessels or shorebased processors

Please select either the PDF or excel version of the survey for the sector or sectors that your company participates in (catcher vessels, catcher-processors, motherships, or shorebased processors). If you have multiple vessels or shorebased processors, fill out **either one PDF survey for each individual vessel and shorebased processor** or **one excel form for the entire fleet**. If you choose to fill out the excel form, please note there are separate excel files for each sector. The PDF form contains all definitions and instructions for the survey while the corresponding excel file contains simplified descriptions and multiple columns for multiple vessels/processors for ease of use for larger companies**. Please refer to the PDF survey for full information about each survey question**.

Step 2: Fill out survey for 'other' employment and spending

If you have other employment or costs that are not associated with individual vessels or shorebased processors, please fill out the "***Survey of Other Employment and Spending***" form. This includes any general and administrative costs associated with fishing and processing businesses not already reported for individual vessels or shorebased processors.

Step 3: Review and submit

Please review your responses to ensure that there is no duplication and that all answers have been answered to the best of your ability. Once you are finished, send all completed forms and excel files as attachments to Melissa Errend at *melissa.errend@norecon.com*, by February 5th, 2025.

Confidentiality Notice

All individual responses are considered confidential. All responses will be aggregated across individuals and sectors in the final products provided to APFA to ensure the confidentiality of responses. Please return signed NDAs to Ron Rogness at *ron.rogness@alaspapollock.org*

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Survey of Catcher Processors

The information requested below is for **individual vessels** that harvested and processed Alaska pollock in 2023. If this vessel is owned and operated by a multi-vessel or vertically integrated company, general and administrative costs and other non-harvesting and non-processing employment can be reported on a separate survey "***Survey of Headquarters and Other Employment and Spending***". Please see the instructions document for further information on the purpose of this survey and instructions for submitting responses.

All individual answers are considered confidential, and all responses will be aggregated to protect the identity of individual companies and respondents.

All questions relate to employment, landings, earnings, and costs for the 2023 calendar year only.

Please complete all questions. If a question is not applicable to you or the vessel you are answering questions about, write "NA" in the answer box**. If an exact number is not available, please estimate to the best of your ability.**

If you have questions on the survey, please reach out to Melissa Errend at *melissa.errend@norecon.com*

1. What is the name of the catcher processor that you are providing answers about? _____

2. What is the vessel's **principal port**? In other words, where does this vessel spend the majority of its time operating out of? (e.g., Dutch Harbor, Kodiak, Seattle, Other): _____

3. What is the name of your company: _____

4. Please provide your contact information in case we have questions about your responses:
    a. Name: _____
    b. Email: _____
    c. Phone Number: _____

The following questions relate to all fisheries and activities the vessel participated in during the **2023 calendar year.**

5. Please estimate the number of **days at sea** you operated this vessel at sea in each of the following fisheries and activities. **For the BSAI Alaska pollock fishery, include days at sea during directed fishing trips for Alaska pollock.** Do not include time spent steaming between the West Coast and Alaska or time spent in port. If you cannot estimate your days at sea across activities, you may put 'NA' and only answer question #6.
   - Alaska pollock BSAI trawl fishery: _____
   - All other Alaska fisheries (including tendering, chartering, or research): _____
   - West coast fisheries (including tendering, chartering, or research: _____

6. Please estimate the **total gross revenue** from each of the following activities that this vessel was engaged in 2023. **For the BSAI Alaska pollock fishery, include revenue from all landed species from directed fishing trips for Alaska pollock**.
   - Alaska pollock BSAI trawl fishery ($): _____
   - All other Alaska fisheries (including tendering, chartering, or research) ($): _____
   - West coast fisheries (including tendering, chartering, or research ($): _____

For the following questions please provide answers about your landings and revenue from your participation in the Alaska pollock fishery **in 2023 only.**

7. Provide the vessel's total landings of **Alaska pollock** (mt) _____

8. Provide the first sales volume of **Alaska pollock** (mt): _____

9. Provide the vessel's total first sales value of **Alaska pollock** ($): _____

10. Provide the percentage of total first sales value of **Alaska pollock** by the state of offload (%):
    - Alaska: _____
    - Washington: _____

Employment

Please provide information about the employment of individuals on this vessel while participating in the Alaska pollock fishery **in 2023.**

Processing crew includes line workers, fishmeal crew, quality control, technicians, cleanup, factory managers, mechanics who work on processing equipment, and combis.

Non-processing crew includes captain, deckhands, wheelhouse, galley, and engine room.

11. Provide the typical number of processing crew positions and the number of non-processing crew positions (including the captain).
    - Number of processing crew positions: _____
    - Number of non-processing crew positions: _____

12. What is the total number of **individuals** (not including observers) who worked on this vessel as processing crew and non-processing crew while **participating in the Alaska pollock fishery**. This value should represent the total number of individuals who worked on this vessel at any point during the year, rather than the number of positions reported in Question 11:
    - Processing crew: _____
    - Non-processing crew: _____

13. How many **processing crew** individuals who worked on this vessel while participating in the Alaska pollock fishery reside in the following states or countries:
    - Alaska: _____
    - Washington: _____
    - Oregon: _____
    - Other US state: _____
    - International Country: _____
14. How many **non-processing crew** individuals who worked on this vessel while participating in the Alaska pollock fishery reside in the following states or countries:
    - Alaska: _____
    - Washington: _____
    - Oregon: _____
    - Other US state: _____
    - International Country: _____

Capital Spending

Please provide total spending **in 2023** on capitalized expenses in the following categories:

Vessel and On-board Equipment: Include the purchase of a new or used vessel, electronics, safety equipment, and machinery not used to harvest or process fish. Also includes any major upgrades, repairs, or maintenance to the vessel or equipment.

Fishing Gear: Include purchase of nets, doors, cables, and fishing machinery used for fishing for pollock. Include any repairs or maintenance to fishing gear. ***Exclude any fishing gear used primarily for other fisheries.***

Processing Equipment: Include any equipment used to process (including freezing) or head and gut fish onboard the vessel, such as freezers, storage equipment and packing equipment, conveyors. ***Exclude all equipment, machines, and buildings based primarily on shore. Exclude processing equipment not used to process Alaska pollock.***

15. Please indicate the total amount ($) and estimate the share of capitalized spending (%) by the location where each item was purchased from (percentages must sum to 100%). Assign the location based on your best estimate of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice. Please round costs to the nearest $1 and round percentages to the nearest 1%:

Category	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Vessel and onboard equipment	$	%	%	%	%	%
Fishing gear	$	%	%	%	%	%
Processing equipment	$	%	%	%	%	%

16. Was any capitalized expenditure used to fish or process fish in other fisheries other than directed fisheries for Alaska pollock? (please check yes or no)
    - Vessel and onboard equipment: ☐ yes ☐ no
    - Fishing gear: ☐ yes ☐ no
    - Processing equipment: ☐ yes ☐ no

Operations and Maintenance (O&M) Spending

Please provide labor and non-labor O&M expenses for this vessel while participating in the Alaska pollock fishery **in 2023.**

17. Please indicate your total labor expenses (include wages, bonuses, benefits, payroll taxes & unemployment insurance) **while participating in the Alaska pollock fishery.** Exclude any payments for revenue adjustments from 2022 or other years**:**
    - Payments to processing crew: _____
    - Payments to non-processing crew: _____

18. Please provide your expenses while participating in the **Alaska pollock fishery** and estimate the proportion of spending by location where each item was purchased (percentages must sum to 100%). Exclude fuel costs or other expenses associated with steaming between the West Coast and Alaska. Assign the location based on your **best estimate** of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice. Please round costs to the nearest $1 and round percentages to the nearest 1%.

Item	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Food	$	%	%	%	%	%
Fuel and lubrication	$	%	%	%	%	%
Bering Sea Pollock observer fees, electronic monitoring, and reporting costs (such as Sea State)*	$	%	%	%	%	%

Note: only Bering Sea Alaska pollock monitoring fees are requested, Gulf of Alaska monitoring fees will be calculated based on reported landings.*

19. Please provide your expenses while participating in **all <u>Alaska</u> fisheries in 2023** (including the Alaska pollock fishery and all other Alaska fisheries, as defined in question 5). For each category, please indicate the total amount ($) and percentage (%) allocated to each state or country. Assign the location based on your **best estimate** of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice. Please round costs to the nearest $1 and round percentages to the nearest 1%.

If you cannot attribute costs to Alaska fisheries only, please provide the total expense amount for each category in in 2023 and check this box: ☐

Item	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Satellite/VMS or other communication	$	%	%	%	%	%
Freight	$	%	%	%	%	%
Travel	$	%	%	%	%	%
Offloading (cross-dock fees, tariffs, hoist fees, etc.)	$	%	%	%	%	%
Fishing association fees	$	%	%	%	%	%
Cooperative fees	$	%	%	%	%	%
State and federal permit/license fees	$	%	%	%	%	%

20. Please provide your **vessel's** total 2023 expenses in the following categories. For each category, please indicate the total amount ($) and percentage (%) allocated to each state or country. Assign the location based on your **best estimate** of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice. Please round costs to the nearest $1 and round percentages to the nearest 1%.

Item	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Lease expenses for vessels and onboard equipment	$	%	%	%	%	%
Repair and maintenance expenditures (e.g. nets, doors, and cables that were fully expensed in 2023)	$	%	%	%	%	%
Insurance associated with the operation of this vessel (not including employee health insurance)	$	%	%	%	%	%
Moorage and wharfage	$	%	%	%	%	%
Management fees (paid to management companies)	$	%	%	%	%	%

Survey of Catcher Vessels

The information requested below is for **individual vessels** that targeted Alaska pollock in 2023. All questions relate to activities, employment, and spending associated with this vessel in 2023. If this vessel is owned and operated by a multi-vessel or vertically integrated company, general and administrative costs and other non-harvesting employment can be reported on a separate survey "***Survey of Headquarters and Other Employment and Spending***". Please see the instructions document for further information on the purpose of this survey and instructions for submitting responses.

All individual answers are considered confidential, and all responses will be aggregated to protect the identity of individual companies and respondents.

All questions relate to employment, landings, earnings, and costs for the 2023 calendar year only.

Please complete all questions. If a question is not applicable to you or the vessel you are answering questions about, write "NA" in the answer box. **If an exact number is not available, please estimate to the best of your ability.**

If you have questions on the survey, please reach out to Melissa Errend at *melissa.errend@norecon.com*

1. What is the name of the catcher vessel that you are providing answers about? _____

2. What is the vessel's **principal port**? In other words, where does this vessel spend the majority of its time operating out of? (e.g., Dutch Harbor, Kodiak, Seattle, Other): _____

3. What is the name of your company? _____

4. Please provide your contact information in case we have questions about your responses:
    a. Name: _____
    b. Email: _____
    c. Phone Number: _____

The following questions relate to all fisheries and activities the vessel participated in during the **2023 calendar year.**

5. Please estimate the number of **days at sea** you operated this vessel in each of the following fisheries and activities. **For the BSAI and GOA Alaska pollock fisheries, include days at sea during directed fishing trips for Alaska pollock.** For commercial fisheries, do not include time spent steaming between the West Coast and Alaska or time spent in port. If you cannot estimate your days at sea across activities, you may put 'NA' and only answer question #6. This information will be used to disaggregate costs to estimate the proportion related to participation in the Alaska pollock fishery.
   - Alaska pollock BSAI trawl fishery: _____
   - Alaska pollock GOA trawl fishery: _____
   - All other Alaska fisheries (including tendering, chartering, or research): _____
   - West coast fisheries (including tendering, chartering, or research: _____

6. Please estimate the **total gross revenue** from each of the following activities that this vessel was engaged in 2023. **For the BSAI and GOA Alaska pollock fisheries, include revenue from all landed species from directed fishing trips for Alaska pollock**. This information will be used to understand the economic impacts of the Alaska pollock fishery and disaggregate costs to estimate the proportion related to participation in the Alaska pollock fishery.
   - Alaska pollock BSAI trawl fishery ($): _____
   - Alaska pollock GOA trawl fishery ($): _____
   - All other Alaska fisheries (including tendering, chartering, or research) ($): _____
   - West coast fisheries (including tendering, chartering, or research ($): _____

For the following questions please provide answers about your landings and revenue from this vessel's directed fishing trips for Alaska pollock **in 2023 only.**

7. Provide the vessel's total landings of **Alaska pollock** from directed fishing trips for Alaska pollock (mt): _____

8. Provide the vessel's total ex-vessel revenue of **Alaska pollock** from directed fishing trips for Alaska pollock ($): _____

Employment

Please provide responses for the number of positions and individuals working on your vessel **in 2023.**

9. Provide the typical crew size (the number of positions, including the captain, not including observers) **on directed fishing trips for Alaska pollock**: _____

10. Provide the total number of **individuals** who worked as crewmembers (including the captain) on this vessel (not including observers) while **on directed fishing trips for Alaska pollock**. This value should represent the total number of individuals who worked as crew on your vessel at any point during the year, rather than the number of positions reported in Question 9: _____

11. How many **individuals** who worked as crewmembers (including the captain) on this vessel (not including observers) while **on directed fishing trips for Alaska pollock** reside in the following states or countries:
    - Alaska: _____
    - Washington: _____
    - Oregon: _____
    - Other US state: _____
    - International Country: _____

Capital Spending

Please provide total spending **in 2023** on capitalized expenses in the following categories:

Vessel and On-board Equipment: Include the purchase of a new or used vessel, electronics, safety equipment, and machinery not used to harvest or process fish. Also includes any major upgrades to the vessel or equipment.

Fishing Gear: Include purchase of nets, doors, cables, and fishing machinery used for fishing for pollock. Include any repairs or maintenance to fishing gear. ***Exclude any fishing gear used primarily for other fisheries.***

Capitalized Repair and Maintenance: include any capitalized repair and maintenance costs to vessel equipment or to the vessel, such as dry docking.

For each category, provide the total amount spent in 2023 as well as your estimate of the state where the purchase occurred. This information is important to identify where economic impacts occur. Especially for the location of purchases, good faith estimates are acceptable if you cannot attribute spending to a particular location.

12. For each category of capitalized spending, please indicate the total amount ($) and estimate the percentage (%) allocated to each state or country. Assign the location based on your best estimate of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice. Please round costs to the nearest $1 and round percentages to the nearest 1%:

Category	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Vessel and onboard equipment	$	%	%	%	%	%
Fishing gear	$	%	%	%	%	%
Capitalized repair and maintenance	$	%	%	%	%	%

13. Was any capitalized expenditure used to fish in other fisheries other than directed fisheries for Alaska pollock? (please mark each with yes or no)
    - Vessel and onboard equipment:
      West Coast fisheries: Yes ☐ No ☐
      Other Alaska fisheries: Yes ☐ No ☐
    - Fishing gear:
      West Coast fisheries: Yes ☐ No ☐
      Other Alaska fisheries: Yes ☐ No ☐
    - Capitalized repair and maintenance:
      West Coast fisheries: Yes ☐ No ☐
      Other Alaska fisheries: Yes ☐ No ☐

Operations and Maintenance (O&M) Spending:

Please indicate your 2023 expenses on labor, operations and maintenance. For non-labor expenses, provide the amount spent in 2023 as well as your estimate of where purchases occurred. This information is important to identify where economic impacts occur. Especially for the location of purchases, good faith estimates are acceptable if you cannot attribute spending to a particular location.

14. Please indicate your total labor expenses including wages, bonuses, benefits, payroll taxes & unemployment insurance for captains and crew **during directed fishing trips for Alaska pollock**. Exclude any payments for revenue adjustments from 2022 or other years:

    - Total payments to captains and crew: _____

15. Please provide your expenses during **directed fishing trips for Alaska pollock**. For each expense category, please indicate the total amount ($) and estimate the percentage (%) allocated to each state or country. Exclude fuel costs or other expenses associated with steaming between the West Coast and Alaska. Assign the location based on your **best estimate** of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice. Please round costs to the nearest $1 and round percentages to the nearest 1%.

Item	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Food						
Fuel and lubrication						
Bering Sea Pollock observer fees, electronic monitoring, and reporting costs (such as Sea State)*						

Note: only Bering Sea Alaska pollock monitoring fees are requested, Gulf of Alaska monitoring fees will be calculated based on reported landings.*

16. Please provide your expenses while participating in **all <u>Alaska</u> fisheries in 2023** (including directed pollock fishing trips and participation in all other Alaska fisheries, as defined in question 6). For each category, please indicate the total amount ($) and percentage (%) allocated to each state or country. Assign the location based on your **best estimate** of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice. Please round costs to the nearest $1 and round percentages to the nearest 1%.

If you cannot attribute costs to Alaska fisheries only, please provide the total expense amount for each category in 2023 and check this box: ☐

Item	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Satellite/VMS or other communication	$					
Freight	$					
Travel	$					
Offloading (cross-dock fees, tariffs, hoist fees, etc.)	$					
Fishing association fees	$					
Cooperative fees	$					
State and federal permit/license fees	$					

17. Please provide your **vessel's** total 2023 expenses in the following categories. For each category, please indicate the total amount ($) and percentage (%) allocated to each state or country. Assign the location based on your **best estimate** of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice. Please round costs to the nearest $1 and round percentages to the nearest 1%.

Item	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Lease expenses for vessels and onboard equipment	$	%	%	%	%	%
Repair and maintenance expenditures (e.g. nets, doors, and cables that were fully expensed in 2023)	$	%	%	%	%	%
Insurance associated with the operation of this vessel (not including employee health insurance)	$	%	%	%	%	%
Moorage and wharfage	$	%	%	%	%	%
Management fees (paid to management companies)	$	%	%	%	%	%

Survey of Motherships

The information requested below is for **individual vessels** that processed Alaska pollock in 2023. If this vessel is owned and operated by a multi-vessel or vertically integrated company, general and administrative costs and other employment can be reported on a separate survey "***Survey of Headquarters and Other Employment and Spending***". Please see the instructions document for further information on the purpose of this survey and instructions for submitting responses.

All individual answers are considered confidential and all responses will be aggregated to protect the identity of individual companies and respondents.

All questions relate to employment, landings, earnings, and costs for the 2023 calendar year only.

Please complete all questions. If a question is not applicable to you or the vessel you are answering questions about, write "NA" in the answer box. **If an exact number is not available, please estimate to the best of your ability.**

If you have questions on the survey, please reach out to Melissa Errend at *melissa.errend@norecon.com*

1. What is the name of the mothership vessel that you are providing answers about? _____

2. What is the vessel's **principal port**? In other words, where does this vessel spend the majority of its time operating out of? (e.g., Dutch Harbor, Kodiak, Seattle, Other): _____

3. What is the name of your company: _____

4. Please provide your contact information in case we have questions about your responses:
    a. Name: _____
    b. Email: _____
    c. Phone Number: _____

The following questions relate to all fisheries and activities the vessel participated in during the **2023 calendar year.**

5. Please estimate the number of **days at sea** you operated this vessel at sea in each of the following fisheries and activities. **For the BSAI Alaska pollock fishery, include days at sea while participating in the Alaska pollock fishery.** Do not include time spent steaming between the West Coast and Alaska or time spent in port. If you cannot estimate your days at sea across activities, you may put 'NA' and only answer question #6:
   - Alaska pollock BSAI trawl fishery: _____
   - Other Alaska fisheries (including tendering, chartering, or research): _____
   - West coast fisheries (including tendering, chartering, or research: _____

6. Please estimate the **total gross revenue** from each of the following activities that this vessel was engaged in 2023. **For the BSAI Alaska pollock fishery, include first sales value from all species purchased from vessels participating in the Alaska pollock fishery**.
   - Alaska pollock BSAI trawl fishery ($): _____
   - All other Alaska fisheries (including tendering, chartering, or research) ($): _____
   - West coast fisheries (including tendering, chartering, or research ($): _____

For the following questions please provide answers about your landings and revenue from your participation in the Alaska pollock fishery **in 2023 only.**

7. Provide the vessel's total weight of **Alaska pollock** received from catcher vessels (mt): _____

8. Provide the total cost of **Alaska pollock** purchased from catcher vessels ($): _____

9. Provide the first sales volume of Alaska **pollock** (mt): _____

10. Provide the vessel's total first sales value of Alaska **pollock ($)**: _____

11. Provide the percentage of total first sales value of Alaska **pollock** by the state of offload (%):
    - Alaska: _____
    - Washington: _____

Employment

Please provide information about the employment of individuals on this vessel while participating in the Alaska pollock fishery **in 2023.**

Processing crew includes line workers, fishmeal crew, quality control, technicians, cleanup, factory managers, mechanics who work on processing equipment, and combis.

Non-processing crew includes captain, deckhands, wheelhouse, galley, and engine room.

12. Provide the typical number of processing crew positions and the number of non-processing crew positions (including the captain):
    - Number of processing crew positions: _____
    - Number of non-processing crew positions: _____

13. What is the total number of **individuals** (not including observers) who worked on this vessel as processing crew and non-processing crew while **participating in the Alaska pollock fishery**. This value should represent the total number of individuals who worked at any point during the year, rather than the number of positions reported in Question 12:
    - Processing crew: _____
    - Non-processing crew: _____

14. How many **processing crew** individuals who worked on this vessel while participating in the Alaska pollock fishery reside in the following states or countries:
    - Alaska: _____
    - Washington: _____
    - Oregon: _____
    - Other US state: _____
    - International Country: _____

15. How many **non-processing crew** individuals who worked on this vessel while participating in the Alaska pollock fishery reside in the following states or countries:
    - Alaska: _____
    - Washington: _____
    - Oregon: _____
    - Other US state: _____
    - International Country: _____

Capital Spending

Please provide total spending **in 2023** on capitalized expenses in the following categories:

Vessel and On-board Equipment: Include the purchase of a new or used vessel, electronics, safety equipment, and machinery not used to process fish. Also includes any major upgrades, repairs, or maintenance to the vessel or equipment.

Processing Equipment: Include any equipment used to process (including freezing) or head and gut fish onboard the vessel, such as freezers, storage equipment and packing equipment, conveyors. ***Exclude all equipment, machines, and buildings based primarily on shore. Exclude processing equipment not used to process Alaska pollock.***

16. Please indicate the total amount ($) and share of capitalized spending (%) by the location where each item was purchased from (percentages must sum to 100%). Assign the location based on your best estimate of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice. Please round costs to the nearest $1 and round percentages to the nearest 1% :

Category	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Vessel and onboard equipment						
Processing equipment						

17. Was any capitalized expenditure used to receive or process fish from other fisheries? (please check yes or no)
    - Vessel and onboard equipment: ☐yes ☐no
    - Processing equipment: ☐yes ☐no

Operations and Maintenance (O&M) Spending:

Please provide labor and non-labor O&M expenses for this vessel while participating in the Alaska pollock fishery **in 2023.**

18. Please indicate your total labor expenses (include wages, bonuses, benefits, payroll taxes & unemployment insurance) **while participating in the Alaska pollock fishery**. Exclude any payments for revenue adjustments from 2022 or other years**:**
    - Payments to processing crew: _____
    - Payments to non-processing crew: _____

19. Please provide your expenses while participating in the **Alaska pollock fishery** and estimate the proportion of spending by location where each item was purchased (percentages must sum to 100%). Exclude fuel costs or other expenses associated with steaming between the West Coast and Alaska. Assign the location based on your best estimate of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice. Please round costs to the nearest $1 and round percentages to the nearest 1%.

Item	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Food	$	%	%	%	%	%
Fuel and lubrication	$	%	%	%	%	%
Bering Sea Pollock observer fees, electronic monitoring, and reporting costs (such as Sea State)*	$	%	%	%	%	%

Note: only Bering Sea Alaska pollock monitoring fees are requested, Gulf of Alaska monitoring fees will be calculated based on reported landings.*

20. Please provide your expenses while participating in **all Alaska fisheries in 2023** (including Alaska pollock and all other Alaska fisheries, as defined in question 5). For each category, please indicate the total amount ($) and percentage (%) allocated to each state or country (percentages must sum to 100%). Assign the location based on your **best estimate** of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice. Please round costs to the nearest $1 and round percentages to the nearest 1%.

If you cannot attribute costs to Alaska fisheries only, please provide total O&M expenses from all fisheries this vessel participated in in 2023 and check this box: ☐

Item	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Satellite/VMS or other communication	$	%	%	%	%	%
Freight	$	%	%	%	%	%
Travel	$	%	%	%	%	%
Offloading (cross-dock fees, tariffs, hoist fees, etc.)	$	%	%	%	%	%
Fishing association fees	$	%	%	%	%	%
Cooperative fees	$	%	%	%	%	%
State and federal permit/license fees	$	%	%	%	%	%

21. Please provide your **vessel's** total 2023 expenses in the following categories. For each category, please indicate the total amount ($) and percentage (%) allocated to each state or country. Assign the location based on your **best estimate** of where the service was provided or the product was purchased. If unsure, you may use the ZIP or postal code from the invoice. Please round costs to the nearest $1 and round percentages to the nearest 1%.

Item	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Lease expenses for vessels and onboard equipment	$	%	%	%	%	%
Repair and maintenance expenditures (e.g. nets, doors, and cables that were fully expensed in 2023)	$	%	%	%	%	%
Insurance associated with the operation of this vessel (not including employee health insurance)	$	%	%	%	%	%
Moorage and wharfage	$	%	%	%	%	%
Management fees (paid to management companies)	$	%	%	%	%	%

Survey of Shorebased Processors

The information requested below is for **individual** shorebased processing plants that processed Alaska pollock in 2023.

If this processor is owned and operated by a multi-processor or vertically integrated company, general and administrative costs and other employment can be reported on a separate survey "***Survey of Headquarters and Other Employment and Spending***". Please see the instructions document for further information on the purpose of this survey and instructions for submitting responses.

All individual answers are considered confidential and all responses will be aggregated to protect the identity of individual companies and respondents.

All questions relate to employment, purchases, production, earnings, and costs for the 2023 calendar year only.

Please complete all questions. If a question is not applicable to you or the plant you are answering questions about write "NA" in the answer box. **If an exact number is not available, please estimate to the best of your ability.**

If you have questions on the survey, please reach out to Melissa Errend at *melissa.errend@norecon.com*

1. What is the name of the processing plant that you are providing answers about? _____

2. What is the physical location of the processing plant that you are providing answers about? _____

3. What is the name of your company? _____

4. Please provide your contact information in case we have questions about your responses:
    a. Name: _____
    b. Email: _____
    c. Phone Number: _____

5. How many days in 2023 did this plant operate: _____

6. How many days in 2023 did this plant process Alaska pollock products: _____

Volume and Value of All Production

The following questions ask about your **total fish purchases and production** in 2023.

Do not include products produced from fish acquired in previous years or any payments received for shipping, handling or storage costs.

7. Total weight of fish purchases (mt): _____

8. Total cost of fish purchases ($): _____

9. Total weight of products produced (mt): _____

10. Total first sales value of products produced ($): _____

Volume and Value of Pollock Purchases and Production

The following questions ask about your **Alaska pollock** purchases in 2023.

Do not include products produced from fish acquired in previous years or any payments received for shipping, handling or storage costs.

11. Total weight of **Alaska pollock** purchases (mt): _____

12. Total cost of **Alaska pollock** purchases ($): _____

13. Total weight of **Alaska pollock** products produced (mt): _____

14. Total first sales value of **Alaska pollock** products produced ($): _____

Employment

The following questions ask for information about the number of employees and their wages in 2023. Include full and part-time employees, contract workers, and temporary employees working at this facility. Information is requested separately for production workers and all other workers.

Production workers: includes all workers at this facility engaged in fabricating, processing, assembling, inspecting, receiving, packing, warehousing, shipping, maintenance, repair, janitorial, product development, or transporting product on-site, up to the level of line-supervisors.

Other workers: includes supervision above the line-supervisor level, and other employees physically employed at this plant not described above.

15. Provide information on the total number of individuals who worked at this facility during 2023 in production activities and all other activities. This represents the number of unique individuals who worked at any point during the year, not number of positions. Please include all full-time, part-time, contract, temporary, and workers on paid leave.

    - Total number of individuals working as production workers: _____
    - Total number of individuals working as other workers: _____

16. How many individual **production workers** who worked at this plant reside in the following states or countries:
    - Alaska: _____
    - Washington: _____
    - Oregon: _____
    - Other US state: _____
    - International Country: _____

17. How many **other** individual employees who worked at this plant reside in the following states or countries:
    - Alaska: _____
    - Washington: _____
    - Oregon: _____
    - Other US state: _____
    - International Country: _____

Capital Spending

Please provide total spending **in 2023** on capitalized expenses in the following categories:

Buildings: capitalized expenditure on buildings (excluding land but including building improvements) at this site.

Machinery and equipment*:* new and used machinery and equipment (only including equipment used to process, transport on-site, or store fish on-site) including processing equipment, transportation equipment and storage equipment at this site.

18. Please provide total spending in 2023 on capitalized expenses for each category listed below and estimate the proportion of each cost to the state where the cost was incurred (percentages must sum to 100%.) Please round costs to the nearest $1 and round percentages to the nearest 1%:

Item	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Buildings	$	%	%	%	%	%
Machinery and equipment	$	%	%	%	%	%

Operations and Maintenance (O&M) Spending

19. Provide the total labor expenses for production workers and all other employees.
Include wages, bonuses, benefits (including housing), payroll taxes, unemployment insurance, and other forms of compensation for all full-time and part-time workers, contract workers, temporary workers, and workers on paid leave.

- Total production worker labor expenses: _____
- Total other employee labor expenses: _____

20. Please indicate your non-labor O&M expenses for each item listed below and estimate the proportion of each cost to the state where the cost was incurred (percentages must sum to 100%.) Please round costs to the nearest $1 and round percentages to the nearest 1%.

Item	Amount ($)	Alaska (%)	Washington (%)	Oregon (%)	Rest of U.S. (%)	Rest of the World (%)
Rent or lease of buildings or other structures (including land)	$	%	%	%	%	%
Rent or lease of processing machinery or equipment	$	%	%	%	%	%
Maintenance and repair of buildings, machinery, or equipment	$	%	%	%	%	%
Utilities (e.g., electricity, gas, water, sewer)	$	%	%	%	%	%
Monitoring costs (shoreside only)	$	%	%	%	%	%
Off-site product freezing and storage	$	%	%	%	%	%
Offload expenses paid to other facilities	$	%	%	%	%	%
Insurance	$	%	%	%	%	%
Taxes (property and excise)	$	%	%	%	%	%
Licensing fees	$	%	%	%	%	%
Communications and office supplies	$	%	%	%	%	%
Freight costs of supplies	$	%	%	%	%	%
Production supplies (e.g., knives, hairnets)	$	%	%	%	%	%
Other supplies (cleaning, additives, etc.)	$	%	%	%	%	%
Other (specify):	$	%	%	%	%	%

Survey of Other Employment and Spending

The information requested below is for individual companies and their employment and spending patterns associated with their **non-harvesting** and **non-processing operations** (e.g., headquarters offices, corporate offices, and regional offices not directly associated with harvesting or processing operations). **Do not report any employment or costs reported separately on individual vessel or shorebased processor surveys.**

All questions relate to employment and spending patterns for the 2023 calendar year only.

All individual answers are considered confidential and all responses will be aggregated to protect the identity of individual companies and respondents.

Please complete all questions. If a question is not applicable to the company you are answering questions about write "NA" in the answer box. If an exact number is not available, please estimate to the best of your ability.

If you have questions on the survey, please reach out to Melissa Errend at *melissa.errend@norecon.com*

1. What is the name of your company? _____

2. What proportion (%) of your company's **total gross revenue** was associated with the harvesting, processing, and sale of Alaska pollock products in 2023? (please round to nearest 1%) _____

Employment

The following questions ask for information about the number of **headquarters, satellite offices, or other locations' employees** and their wages in 2023. Include full and part-time employees, contract workers, and temporary employees working at your company's headquarters, satellite offices, or other locations. This excludes employees on individual harvesting vessels, processing vessels or who work at processing plants.

Here, **other employees** include the following:

- Leadership (CEO, COO, CFO, President, Vice-President)
- General counsel and legal
- Public affairs and communications
- Government affairs
- Finance and accounting
- Sales and marketing
- Product development
- Supply chain and logistics
- Customer service
- Business development
- Administrative staff
- Information Technology (IT) staff
- Other staff not on harvesting vessels, processing vessels, or at processing plants owned by your company

3. Provide the total number of individuals who worked as **other employees** in 2023. This represents the number of individuals who worked at any point during the year, not number of positions. Please include all full-time, part-time, contract, temporary, and workers on paid leave.

   ➢ Total number of individuals working as other employees: _____

4. Provide the total labor expenses for **other employees**. Include wages, bonuses, benefits (including housing), payroll taxes, unemployment insurance, and other forms of compensation for all full-time and part-time workers, contract workers, temporary workers, and workers on paid leave.

   ➢ Total other employees labor expenses: _____

5. How many **other employees** reside in the following states or countries:

   ➢ Alaska: _____
   ➢ Washington: _____
   ➢ Oregon: _____
   ➢ Other US state: _____
   ➢ International Country: _____

Capital Spending

Please provide total spending **in 2023** on capitalized expenses not specific to any active vessel or processing plant.

6. Please provide total spending in 2023 on capitalized expenses and estimate the proportion of each cost to the state where the cost was incurred (percentages must sum to 100%.) Please round costs to the nearest $1 and round percentages to the nearest 1%:

Item	Amount ($)	Alaska (%)	Washington (%)	Rest of U.S. (%)	Rest of the World (%)
Total Capital Expenditures	$	%	%	%	%

Operations and Maintenance (O&M) Spending

7. Please indicate your non-labor O&M expenses for each item listed below and estimate the proportion of each cost to the state where the cost was incurred (percentages must sum to 100%.) Do not include any costs directly associated with individual vessels or processing plants. Please round costs to the nearest $1 and round percentages to the nearest 1%.
   Operations and maintenance costs include: rent or lease of buildings and equipment, utilities, insurance, communications and office supplies, building maintenance and repair, or other operating and maintenance costs.

Item	Amount ($)	Alaska (%)	Washington (%)	Rest of U.S. (%)	Rest of the World (%)
Operations and Maintenance Costs (excluding property and excise taxes)	$	%	%	%	%
Property and Excise Taxes	$	%	%	%	%